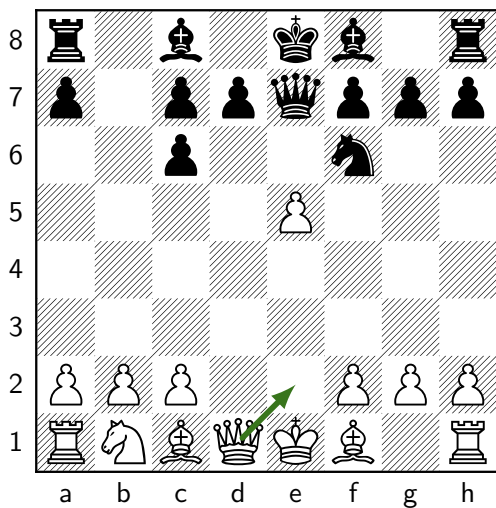


# Melanie-BOTS (CCRL engines) — Scotch Game

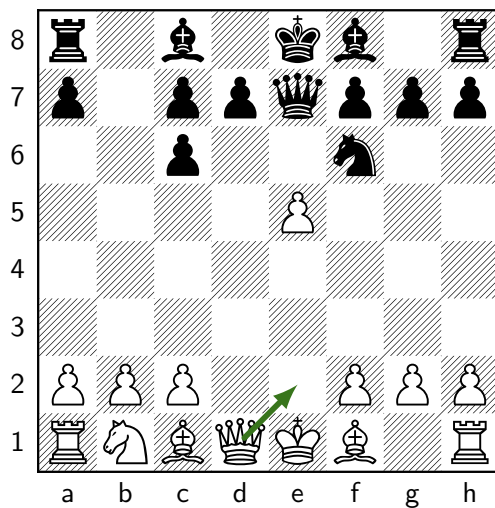
Melanie (Deep\_Vision)

## Abstract

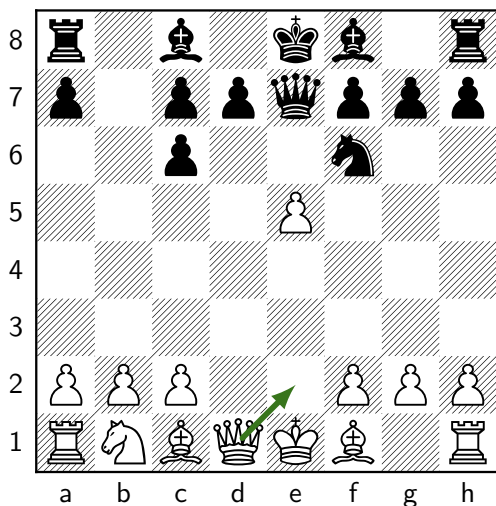
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under  $\mu$  – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



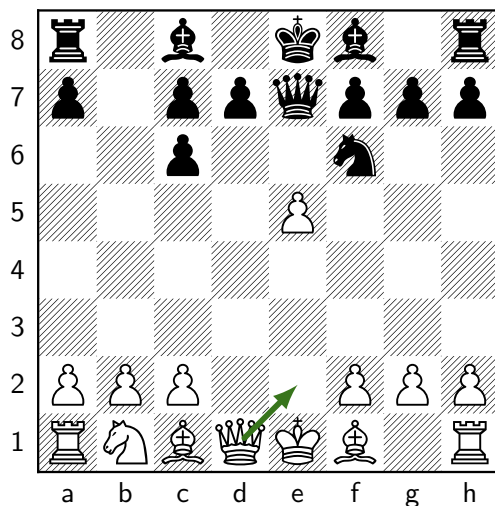
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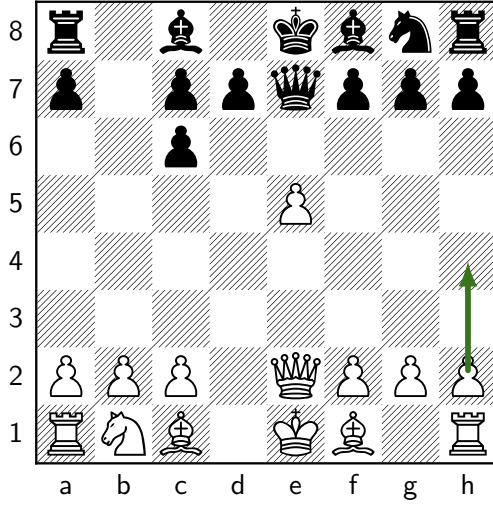


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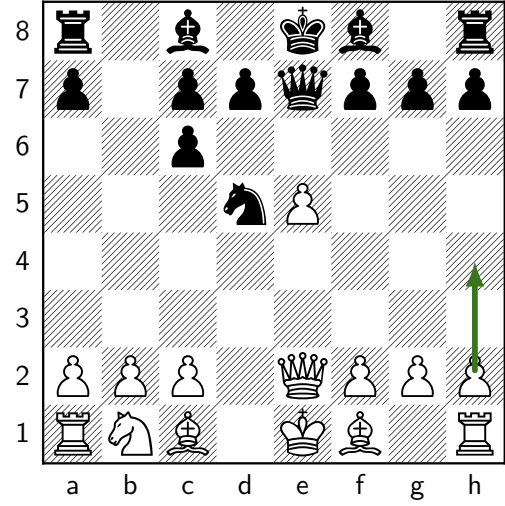


(d)

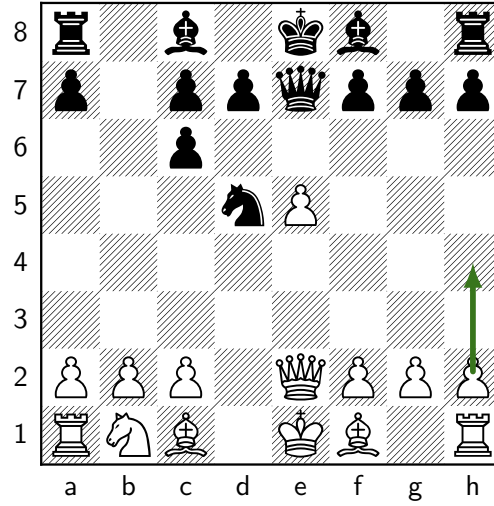
Figure 1: **Move 7, White: Qe2**. (a) Our position; Melanie plays Qe2 (d1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3924` (ply 12) [structural distance  $d=0$ , QAP obj=104]: the master played Qe2; under  $\mu$  our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pf2 as their Nb1, our Pc2 as their Bc1; transferred move d1e2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3925` (ply 12) [structural distance  $d=0$ , QAP obj=104]: the master played Qe2; under  $\mu$  our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pf2 as their Nb1, our Pc2 as their Bc1; transferred move d1e2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#733` (ply 12) [structural distance  $d=0$ , QAP obj=104]: the master played Qe2; under  $\mu$  our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pf2 as their Nb1, our Pc2 as their Bc1; transferred move d1e2 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Qe2 (summed QAP objective 1661, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



(a)

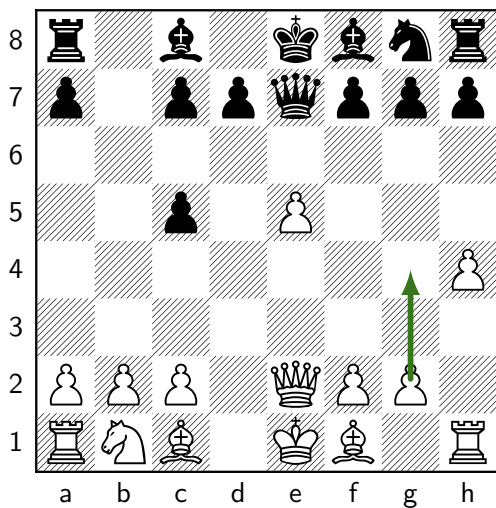


(b)

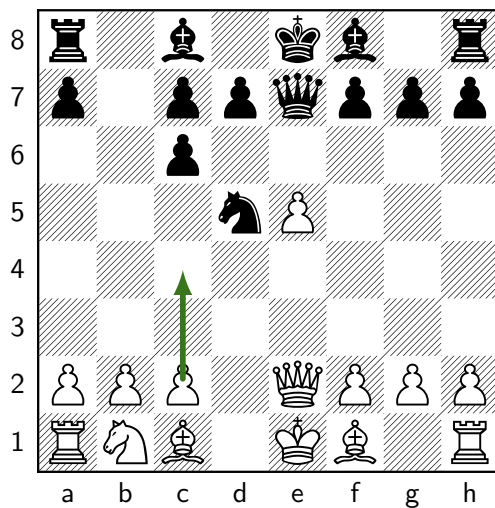


(c)

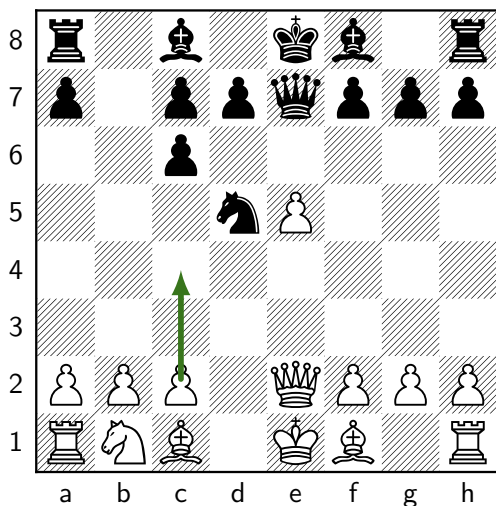
Figure 2: **Move 8, White: h4.** (a) Our position; Melanie plays h4 (h2h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1131` (ply 14) [structural distance  $d=12$ , QAP obj=61]: the master played h4; under  $\mu$  our *Ph2* plays the role of that game’s *Ph2*; cross-type roles: our *Nb1* as their *Ra1*, our *Pf2* as their *Nb1*; transferred move h2h4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1132` (ply 14) [structural distance  $d=12$ , QAP obj=61]: the master played h4; under  $\mu$  our *Ph2* plays the role of that game’s *Ph2*; cross-type roles: our *Nb1* as their *Ra1*, our *Pf2* as their *Nb1*; transferred move h2h4 (**endorses this move**). Synthesis: 2 retrieved analogs transferred to h4 (summed QAP objective 121, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



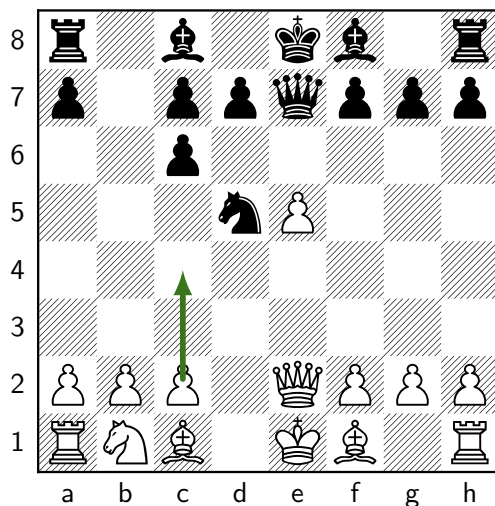
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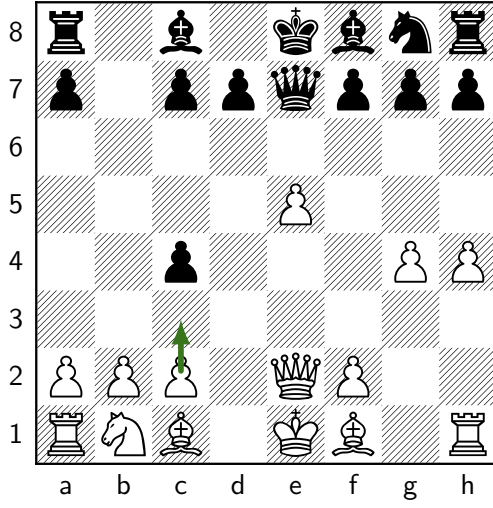


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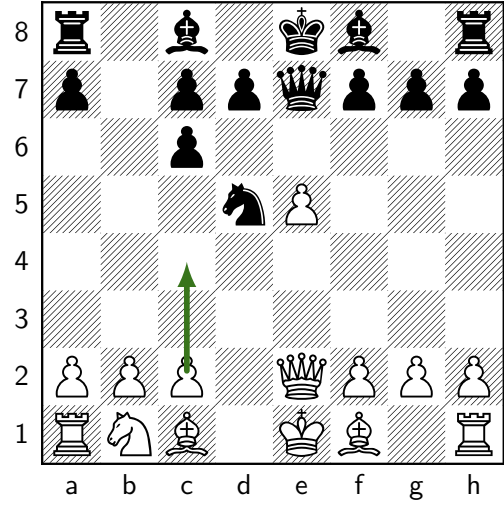


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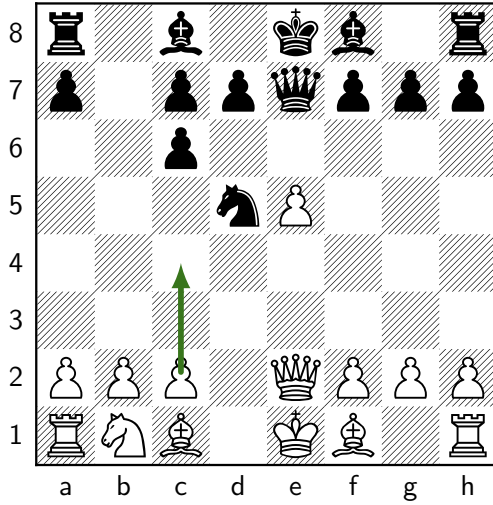
Figure 3: **Move 9, White: g4.** (a) Our position; Melanie plays **g4** (**g2g4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3924` (ply 14) [structural distance  $d=23$ , QAP obj=60]: the master played **c4**; under  $\mu$  our *Pg2* plays the role of that game’s *Pc2*; cross-type roles: our *ra8* as their *Nb1*, our *Rh1* as their *Bc1*; transferred move **g2g4** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3925` (ply 14) [structural distance  $d=23$ , QAP obj=60]: the master played **c4**; under  $\mu$  our *Pg2* plays the role of that game’s *Pc2*; cross-type roles: our *ra8* as their *Nb1*, our *Rh1* as their *Bc1*; transferred move **g2g4** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#733` (ply 14) [structural distance  $d=23$ , QAP obj=60]: the master played **c4**; under  $\mu$  our *Pg2* plays the role of that game’s *Pc2*; cross-type roles: our *ra8* as their *Nb1*, our *Rh1* as their *Bc1*; transferred move **g2g4** (**endorses this move**). Synthesis: 14 retrieved analogs transferred to **g4** (summed QAP objective 843, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



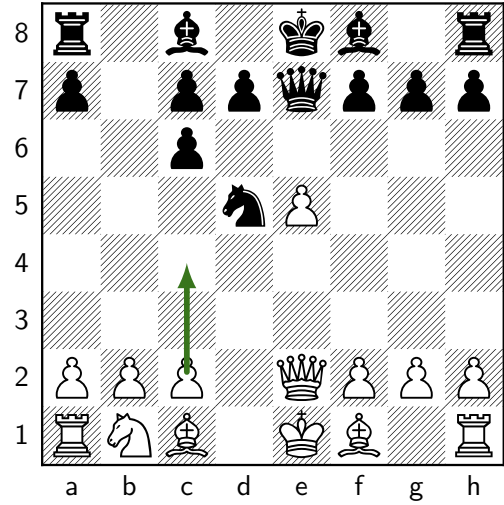
(a)



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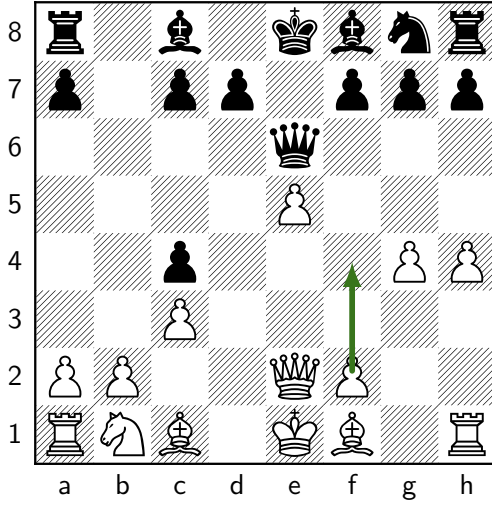


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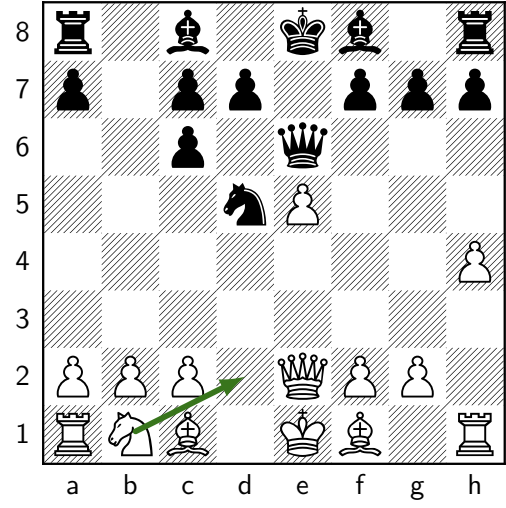


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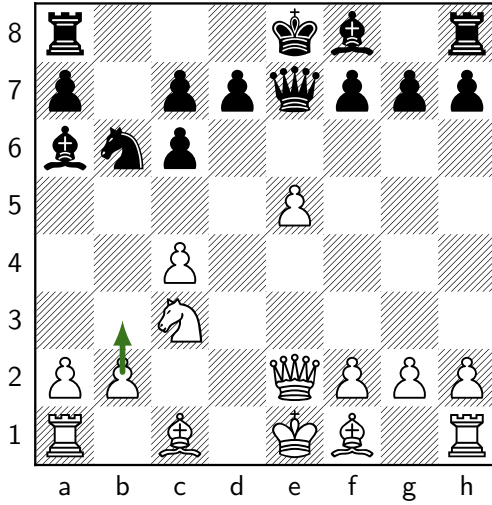
Figure 4: **Move 10, White: c3**. (a) Our position; Melanie plays c3 (c2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3924` (ply 14) [structural distance  $d=42$ , QAP obj=60]: the master played c4; under  $\mu$  our  $Pc2$  plays the role of that game’s  $Pc2$ ; cross-type roles: our  $Nb1$  as their  $Ra1$ , our  $pa7$  as their  $Nb1$ ; transferred move c2c3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3925` (ply 14) [structural distance  $d=42$ , QAP obj=60]: the master played c4; under  $\mu$  our  $Pc2$  plays the role of that game’s  $Pc2$ ; cross-type roles: our  $Nb1$  as their  $Ra1$ , our  $pa7$  as their  $Nb1$ ; transferred move c2c3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#733` (ply 14) [structural distance  $d=42$ , QAP obj=60]: the master played c4; under  $\mu$  our  $Pc2$  plays the role of that game’s  $Pc2$ ; cross-type roles: our  $Nb1$  as their  $Ra1$ , our  $pa7$  as their  $Nb1$ ; transferred move c2c3 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to c3 (summed QAP objective 838, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



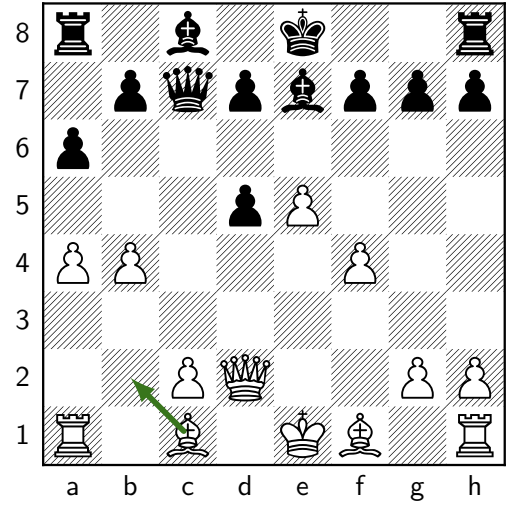
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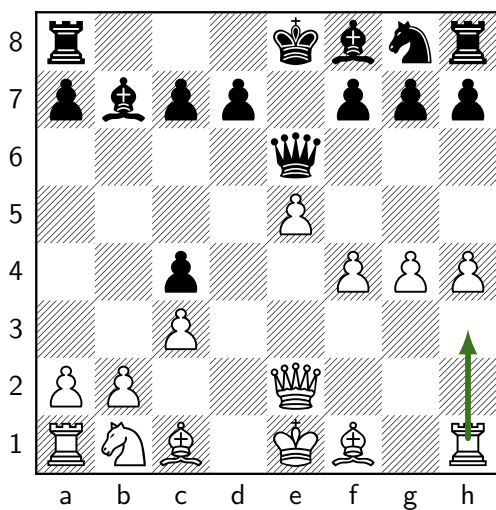


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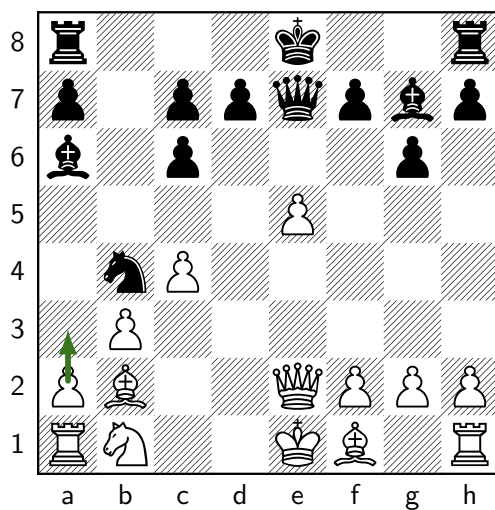


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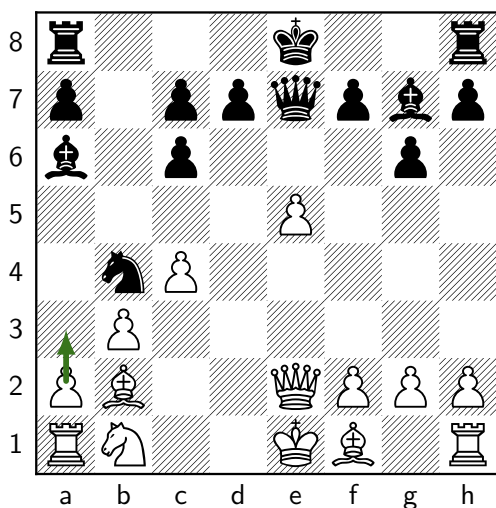
Figure 5: **Move 11, White: f4 (f2f4).** (a) Our position; Melanie plays f4 (f2f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1131` (ply 16) [structural distance  $d=53$ , QAP obj=58]: the master played Nd2; under  $\mu$  our Pf2 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Pc3 as their Bc1, our Ph4 as their Rh1; transferred move f2f4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3924` (ply 18) [structural distance  $d=55$ , QAP obj=54]: the master played b3; under  $\mu$  our Pf2 plays the role of that game’s Pb2; cross-type roles: our ng8 as their Ra1, our Rh1 as their Bc1; transferred move f2f3. (d) Retrieved from `ccrl_top_p0003#1890` (ply 24) [structural distance  $d=47$ , QAP obj=42]: the master played Bb2; under  $\mu$  our Bf1 plays the role of that game’s Bc1; cross-type roles: our Nb1 as their Bf1, our Bc1 as their Rh1; transferred move f1g2. Synthesis: 1 retrieved analog transferred to f4 (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



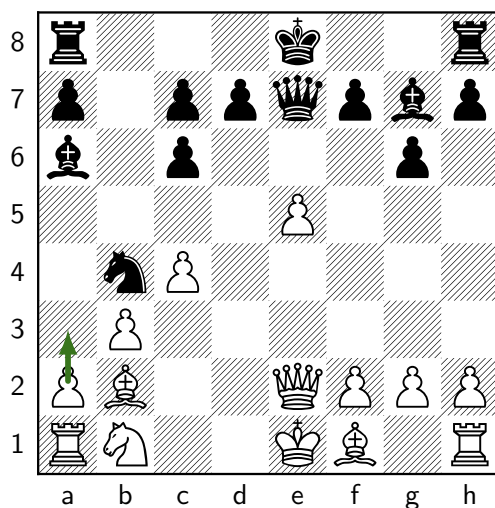
(a)



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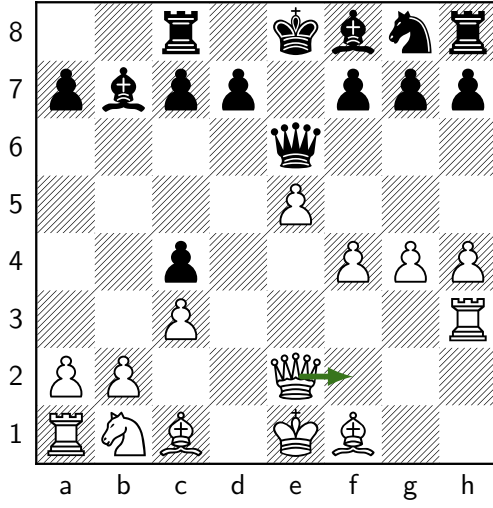


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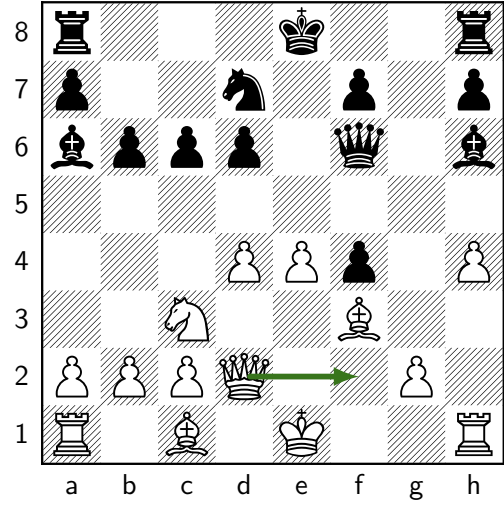


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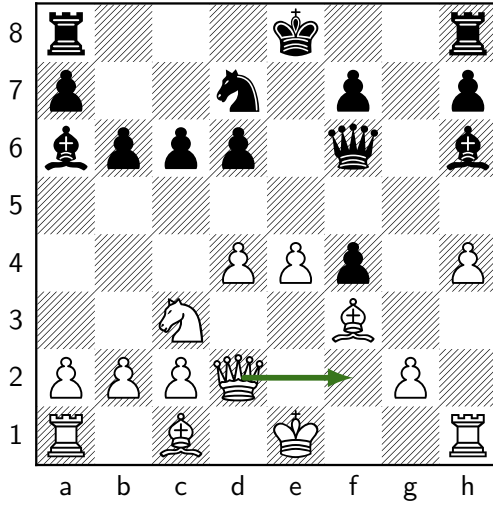
Figure 6: **Move 12, White: Rh3.** (a) Our position; Melanie plays Rh3 (**h1h3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1432` (ply 22) [structural distance  $d=65$ , QAP obj=74]: the master played **a3**; under  $\mu$  our *Rh1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *Pf4* as their *Bb2*, our *bb7* as their *nb4*; transferred move **h1h3** (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1433` (ply 22) [structural distance  $d=65$ , QAP obj=74]: the master played **a3**; under  $\mu$  our *Rh1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *Pf4* as their *Bb2*, our *bb7* as their *nb4*; transferred move **h1h3** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2657` (ply 22) [structural distance  $d=65$ , QAP obj=74]: the master played **a3**; under  $\mu$  our *Rh1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *Pf4* as their *Bb2*, our *bb7* as their *nb4*; transferred move **h1h3** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **Rh3** (summed QAP objective 223, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



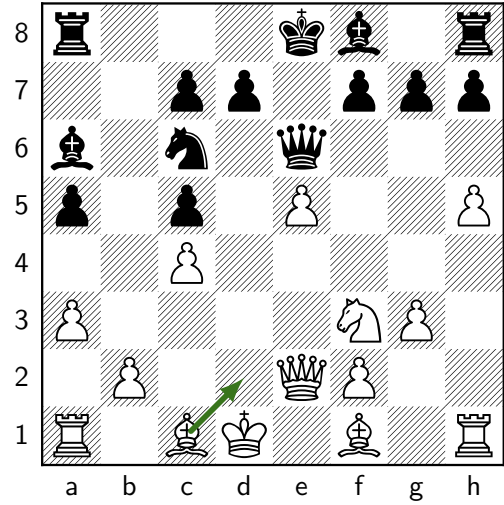
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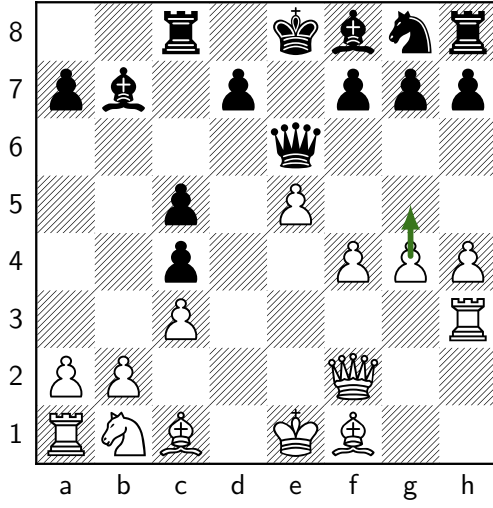
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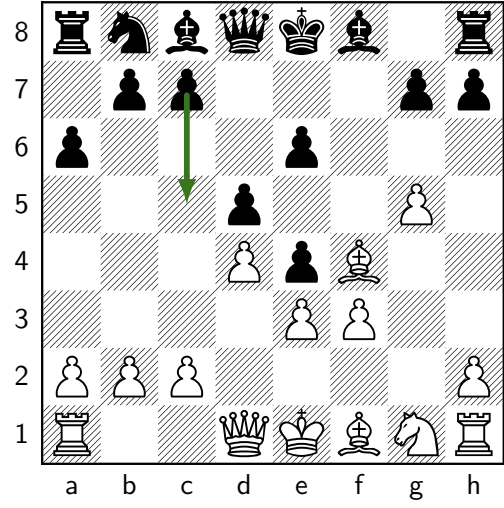
(d)

Figure 7: **Move 13, White: Qf2.** (a) Our position; Melanie plays Qf2 (e2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#482` (ply 24) [structural distance  $d=67$ , QAP obj=71]: the master played Qf2; under  $\mu$  our Qe2 plays the role of that game's Qd2; cross-type roles: our Bc1 as their Ra1, our bf8 as their Rh1; transferred move e2f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1760` (ply 24) [structural distance  $d=67$ , QAP obj=71]: the master played Qf2; under  $\mu$  our Qe2 plays the role of that game's Qd2; cross-type roles: our Bc1 as their Ra1, our bf8 as their Rh1; transferred move e2f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1131` (ply 30) [structural distance  $d=76$ , QAP obj=110]: the master played Bd2; under  $\mu$  our Bf1 plays the role of that game's Bc1; cross-type roles: our Pb2 as their Ra1, our Pf4 as their Bf1; transferred move f1g2. Synthesis: 2 retrieved analogs transferred to Qf2 (summed QAP objective 141, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

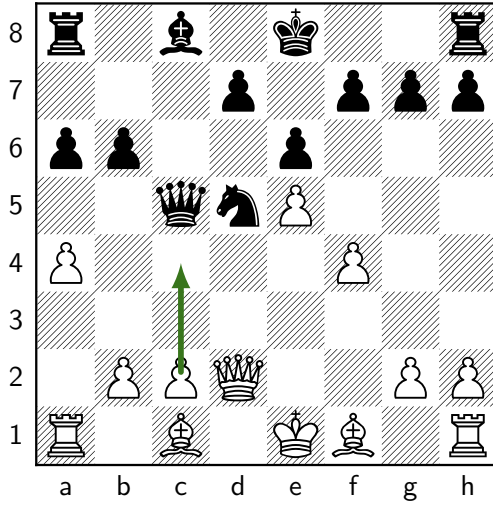




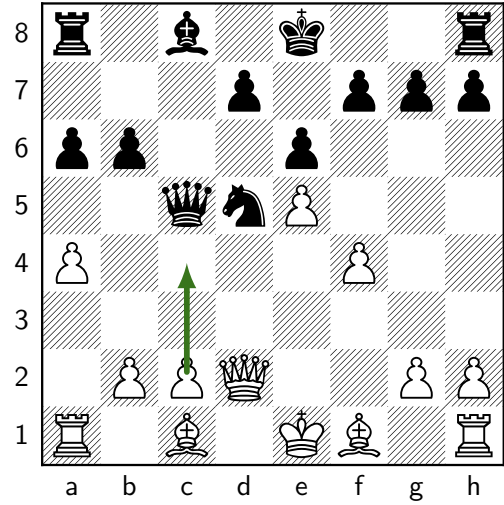
(a)



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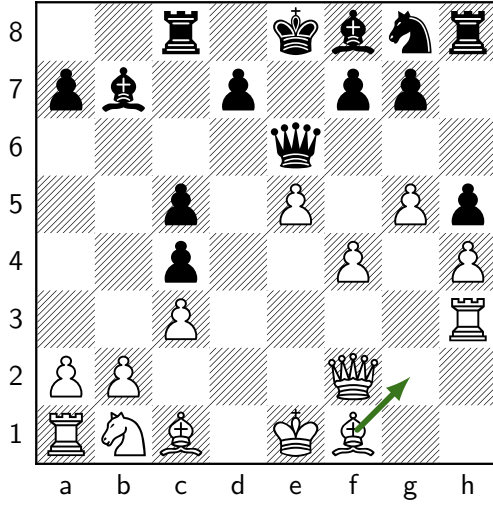


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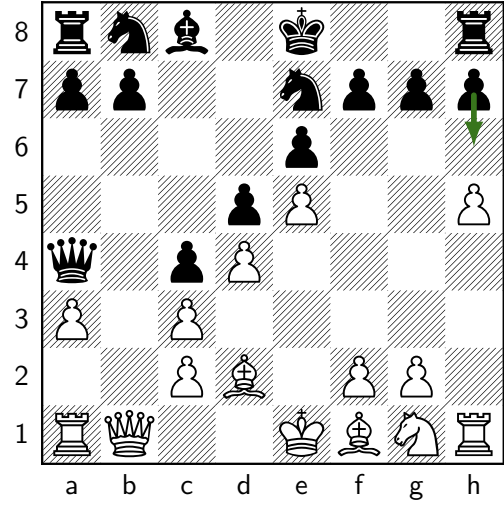


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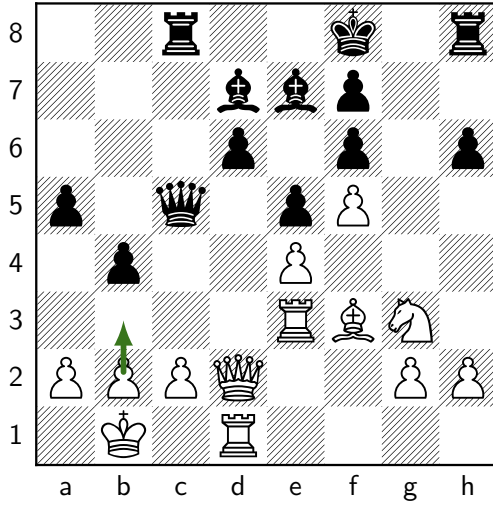
Figure 8: **Move 14, White: g5**. (a) Our position; Melanie plays g5 (g4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3211` (ply 15) [structural distance  $d=72$ , QAP obj=40]: the master played c5; under  $\mu$  our  $Pg4$  plays the role of that game’s  $Pc2$ ; cross-type roles: our  $pc4$  as their  $Nb8$ , our  $Pb2$  as their  $Bc8$ ; transferred move g4g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2212` (ply 24) [structural distance  $d=76$ , QAP obj=32]: the master played c4; under  $\mu$  our  $Pg4$  plays the role of that game’s  $Pc2$ ; cross-type roles: our  $Qf2$  as their  $Ra1$ , our  $pd7$  as their  $Bc1$ ; transferred move g4g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1891` (ply 24) [structural distance  $d=76$ , QAP obj=32]: the master played c4; under  $\mu$  our  $Pg4$  plays the role of that game’s  $Pc2$ ; cross-type roles: our  $Qf2$  as their  $Ra1$ , our  $pd7$  as their  $Bc1$ ; transferred move g4g5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to g5 (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



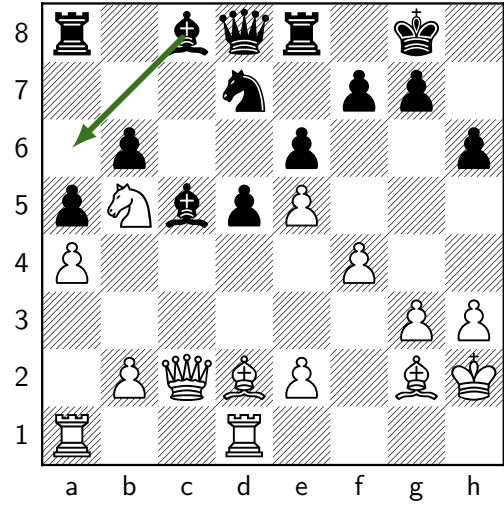
(a)



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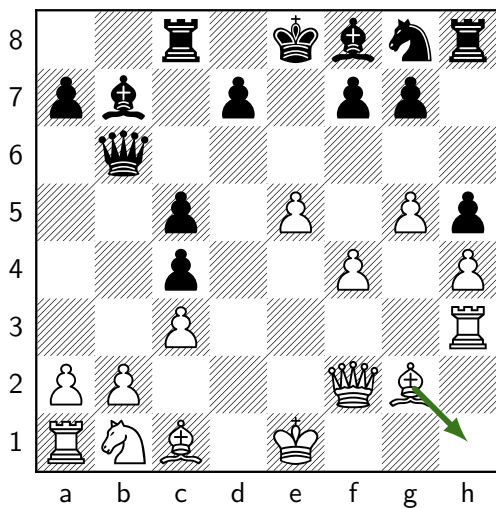


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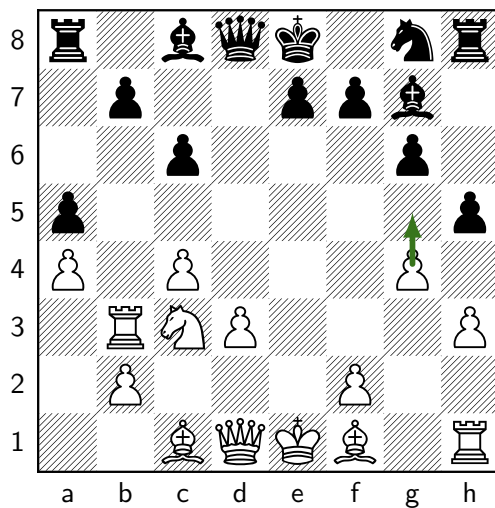


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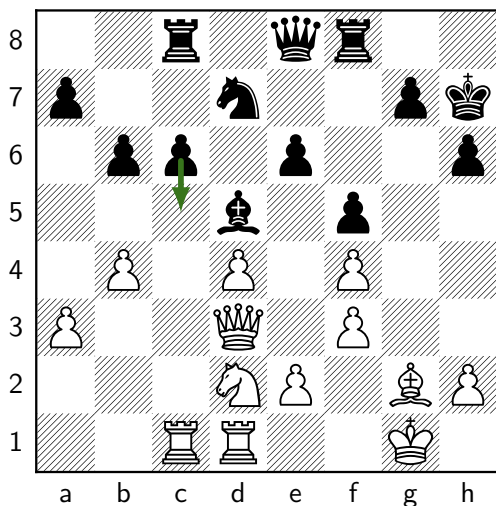
Figure 9: **Move 15, White: Bg2.** (a) Our position; Melanie plays Bg2 (f1g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1581` (ply 19) [structural distance  $d=88$ , QAP obj=48]: the master played h6; under  $\mu$  our Bf1 plays the role of that game’s Ph2 (a **cross-type** correspondence); cross-type roles: our ph5 as their Ra8, our Qf2 as their Bc8; transferred move f1g2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1719` (ply 40) [structural distance  $d=95$ , QAP obj=21]: the master played b3; under  $\mu$  our Bf1 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Qf2 as their Rd1, our Ra1 as their Pa2; transferred move f1g2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#720` (ply 37) [structural distance  $d=97$ , QAP obj=18]: the master played Ba6; under  $\mu$  our Bf1 plays the role of that game’s Bc1; cross-type roles: our Pg5 as their Ra8, our Pb2 as their Qd8; transferred move f1g2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bg2 (summed QAP objective 87, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



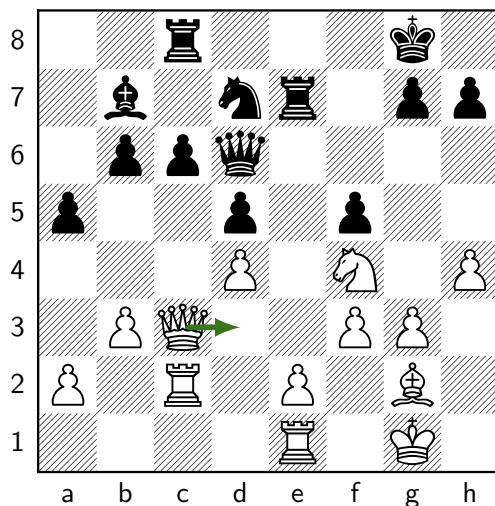
(a)



(b)

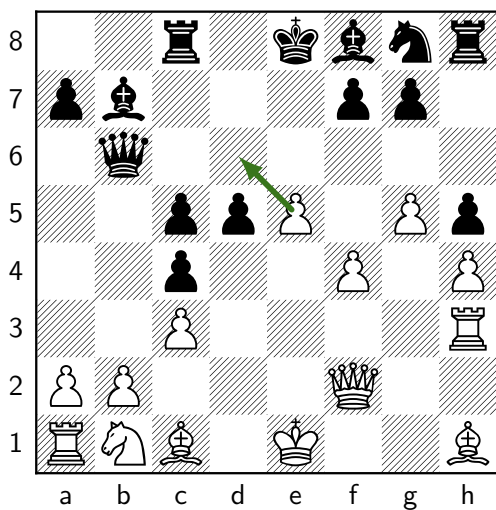


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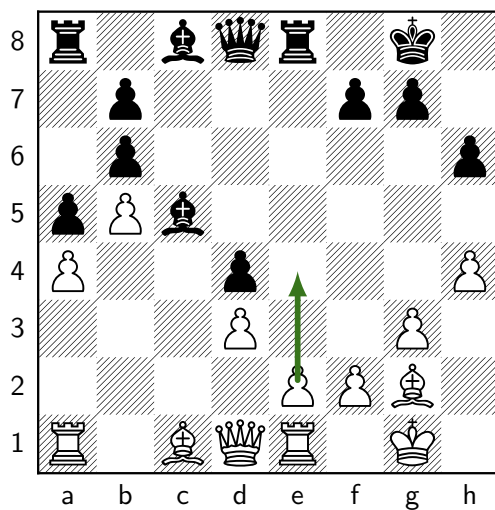


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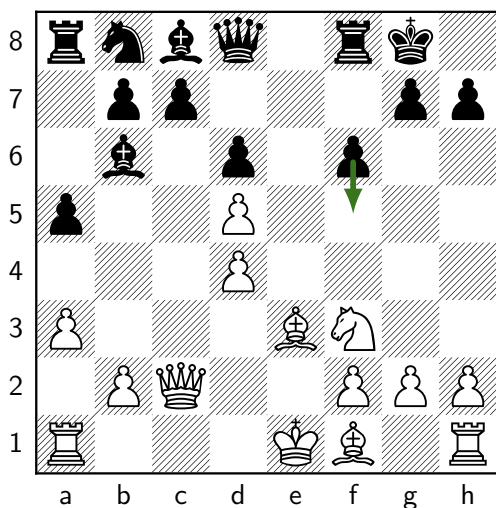
Figure 10: **Move 16, White: Bh1**. (a) Our position; Melanie plays Bh1 (g2h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2833` (ply 24) [structural distance  $d=97$ , QAP obj=48]: the master played g5; under  $\mu$  our *Bg2* plays the role of that game’s *Pg4* (a **cross-type** correspondence); cross-type roles: our *Pf4* as their *Bc1*, our *Nb1* as their *Bf1*; transferred move g2h1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#656` (ply 37) [structural distance  $d=102$ , QAP obj=63]: the master played c5; under  $\mu$  our *Pb2* plays the role of that game’s *Pc3*; cross-type roles: our *Bc1* as their *Rc8*, our *ng8* as their *Qe8*; transferred move b2b3. (d) Retrieved from `ccr1_top_p0003#2182` (ply 42) [structural distance  $d=99$ , QAP obj=54]: the master played Qd3; under  $\mu$  our *Qf2* plays the role of that game’s *Qc3*; cross-type roles: our *Bc1* as their *Re1*, our *Pf4* as their *Rc2*; transferred move f2e3. Synthesis: 2 retrieved analogs transferred to Bh1 (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



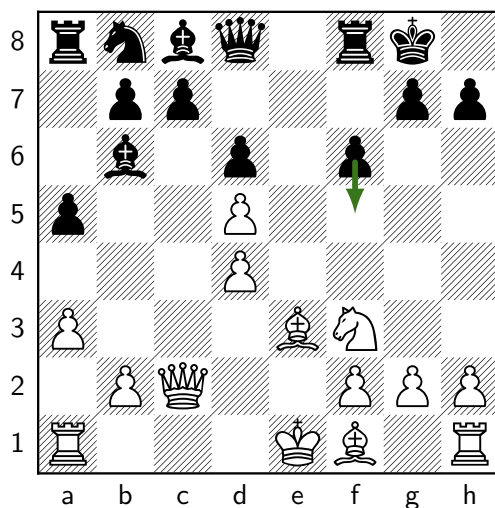
(a)



(b)

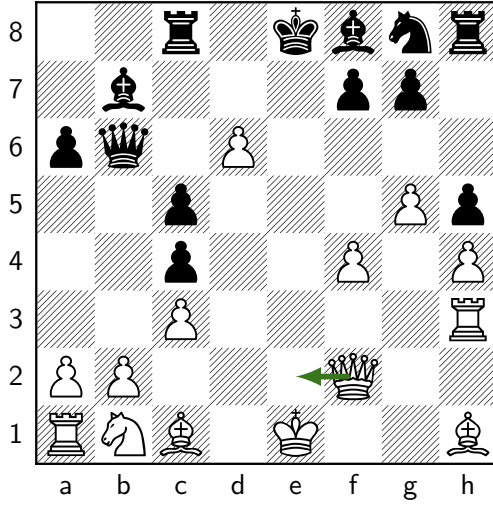


(c)

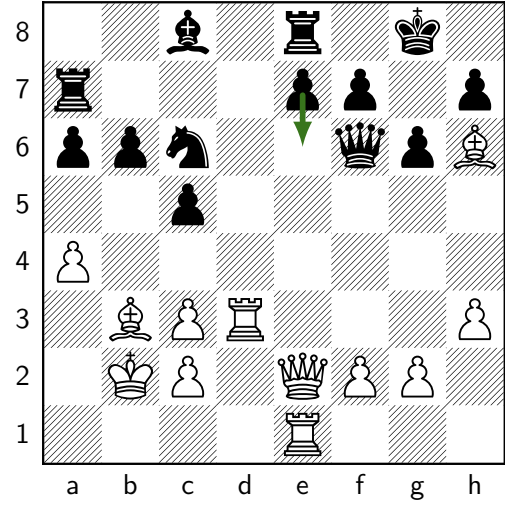


(d)

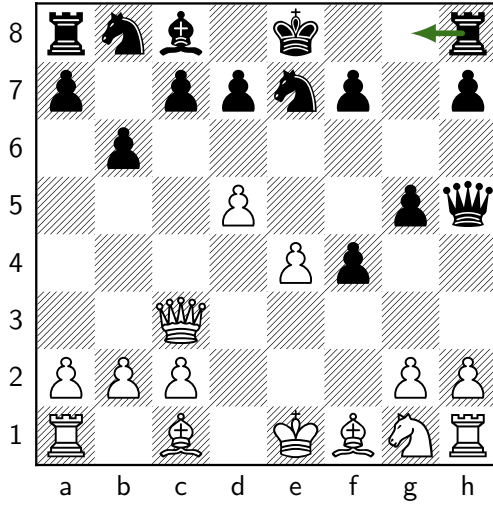
Figure 11: **Move 17, White: exd6.** (a) Our position; Melanie plays exd6 (e5d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#935` (ply 34) [structural distance  $d=98$ , QAP obj=65]: the master played e4; under  $\mu$  our  $Pb2$  plays the role of that game’s  $Pe2$ ; cross-type roles: our  $ph5$  as their  $Ra1$ , our  $Pg5$  as their  $Bc1$ ; transferred move b2b4. (c) Retrieved from `ccr1_top_p0001#245` (ply 23) [structural distance  $d=106$ , QAP obj=61]: the master played f5; under  $\mu$  our  $Pe5$  plays the role of that game’s  $Pf3$ ; cross-type roles: our  $Pc3$  as their  $Nb8$ , our  $Pg5$  as their  $Bc8$ ; transferred move e5e6. (d) Retrieved from `ccr1_top_p0001#1427` (ply 23) [structural distance  $d=106$ , QAP obj=61]: the master played f5; under  $\mu$  our  $Pe5$  plays the role of that game’s  $Pf3$ ; cross-type roles: our  $Pc3$  as their  $Nb8$ , our  $Pg5$  as their  $Bc8$ ; transferred move e5e6. Synthesis: no retrieved analog transferred to exd6 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



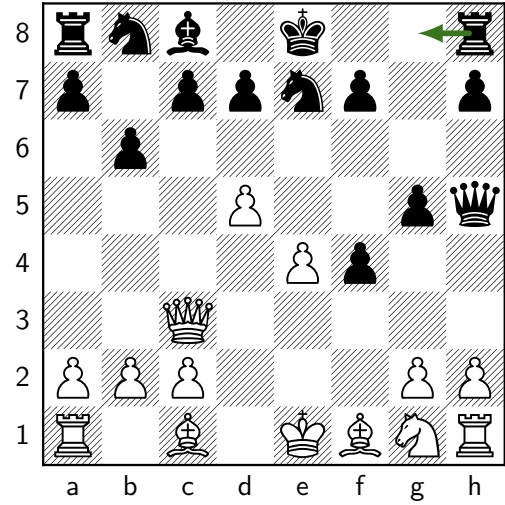
(a)



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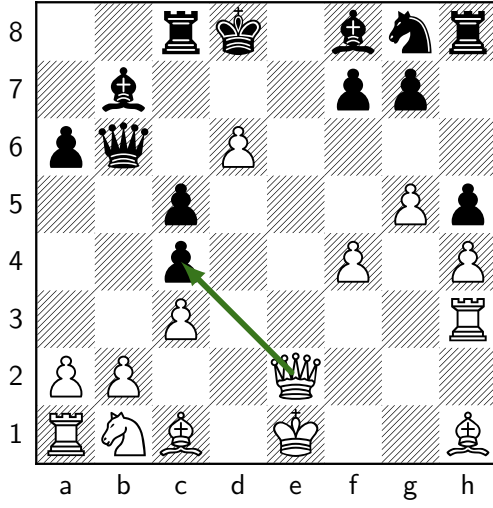


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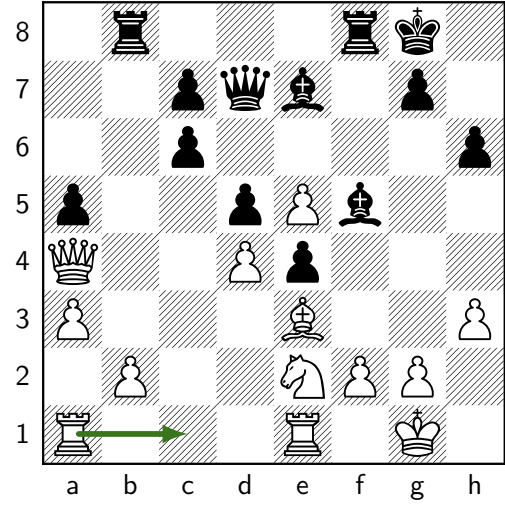


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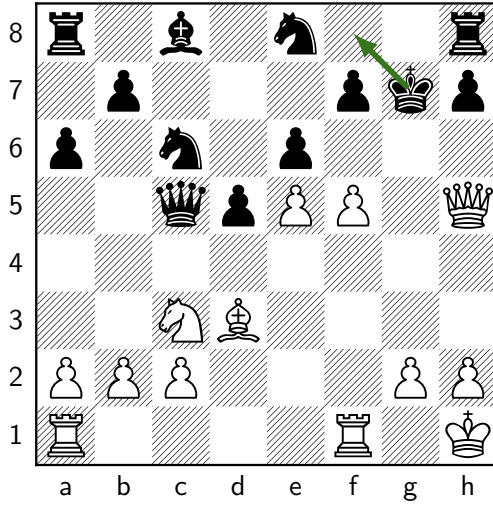
Figure 12: **Move 18, White: Qe2+.** (a) Our position; Melanie plays Qe2+ (f2e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3447` (ply 43) [structural distance  $d=96$ , QAP obj=95]: the master played e6; under  $\mu$  our *Pb2* plays the role of that game's *Pe2*; cross-type roles: our *Pa2* as their *Re8*, our *Pd6* as their *Ra7*; transferred move b2b3. (c) Retrieved from `ccrl_top_p0000#3958` (ply 19) [structural distance  $d=86$ , QAP obj=69]: the master played Rg8; under  $\mu$  our *Pd6* plays the role of that game's *Rh1* (a **cross-type** correspondence); cross-type roles: our *Pc3* as their *Ra8*, our *bb7* as their *Ne7*; transferred move d6d7. (d) Retrieved from `ccrl_top_p0000#3959` (ply 19) [structural distance  $d=86$ , QAP obj=69]: the master played Rg8; under  $\mu$  our *Pd6* plays the role of that game's *Rh1* (a **cross-type** correspondence); cross-type roles: our *Pc3* as their *Ra8*, our *bb7* as their *Ne7*; transferred move d6d7. Synthesis: no retrieved analog transferred to Qe2+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



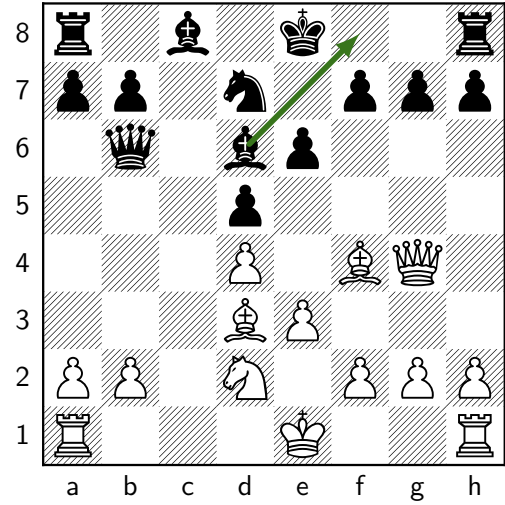
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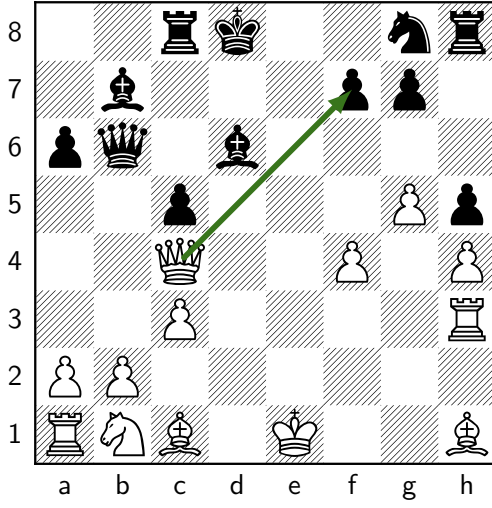


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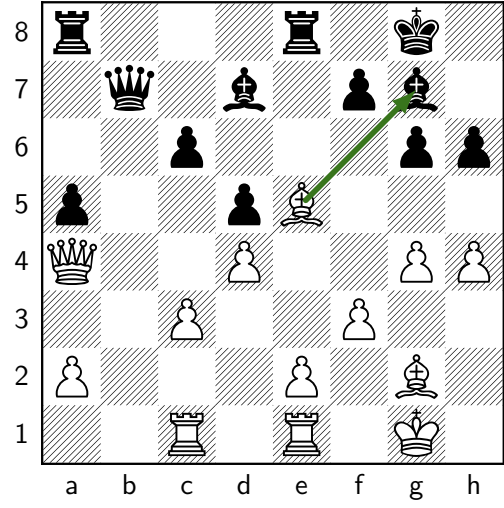


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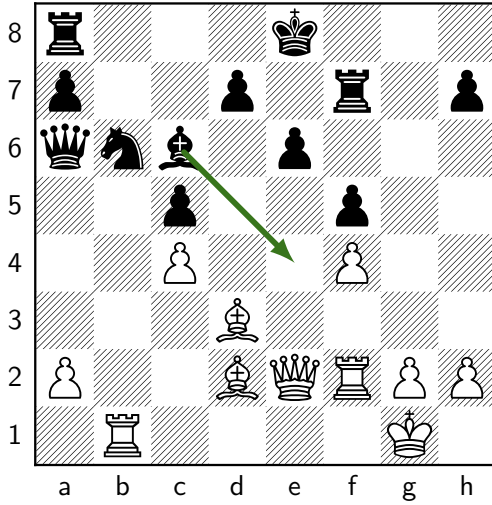
Figure 13: **Move 19, White: Qxc4**. (a) Our position; Melanie plays Qxc4 (e2c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3931` (ply 44) [structural distance  $d=102$ , QAP obj=51]: the master played Rac1; under  $\mu$  our Nb1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Pf4 as their Re1, our Pc3 as their Ne2; transferred move b1a3. (c) Retrieved from `ccr1_top_p0000#3967` (ply 31) [structural distance  $d=99$ , QAP obj=48]: the master played Kf8; under  $\mu$  our Ke1 plays the role of that game’s Kg2; cross-type roles: our pf7 as their Ra8, our Rh3 as their Bc8; transferred move e1f1. (d) Retrieved from `ccr1_top_p0000#3483` (ply 21) [structural distance  $d=102$ , QAP obj=38]: the master played Bf8; under  $\mu$  our Bh1 plays the role of that game’s Bd3; cross-type roles: our Qe2 as their Ra8, our bf8 as their Rh8; transferred move h1f3. Synthesis: no retrieved analog transferred to Qxc4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



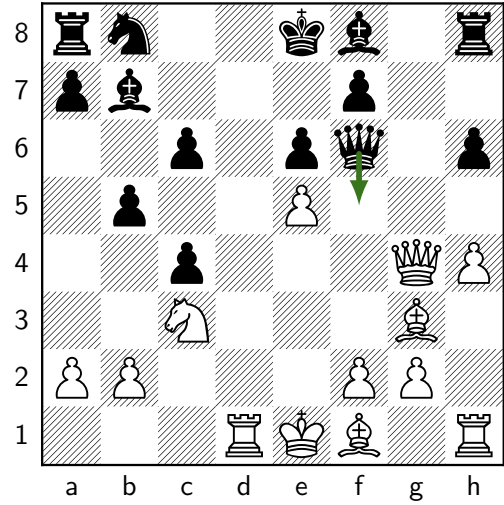
(a)



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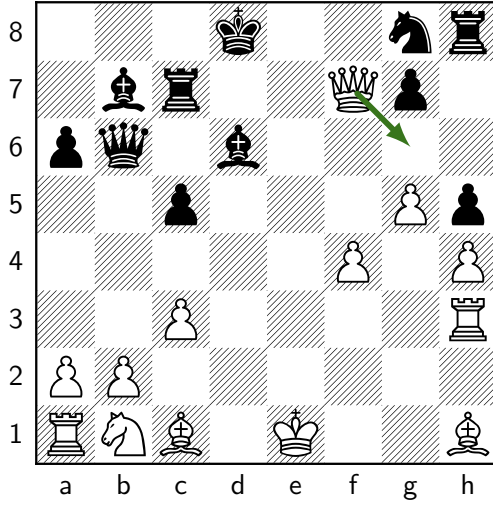
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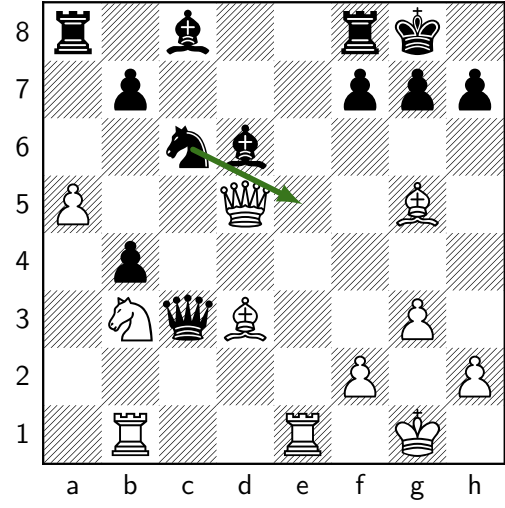
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Figure 14: **Move 20, White: Qxf7.** (a) Our position; Melanie plays Qxf7 (c4f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#276` (ply 40) [structural distance  $d=98$ , QAP obj=87]: the master played Bxg7; under  $\mu$  our Bb1 plays the role of that game’s Be5; cross-type roles: our Nb1 as their Rc1, our ng8 as their Re1; transferred move h1b7. (c) Retrieved from `ccr1_top_p0001#534` (ply 51) [structural distance  $d=99$ , QAP obj=68]: the master played Be4; under  $\mu$  our Pg5 plays the role of that game’s Bc3 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ra8, our ng8 as their Ph7; transferred move g5g6. (d) Retrieved from `ccr1_top_p0003#2087` (ply 27) [structural distance  $d=109$ , QAP obj=57]: the master played Qf5; under  $\mu$  our Qc4 plays the role of that game’s Qf3; cross-type roles: our rc8 as their Nb8, our Nb1 as their Bf8; transferred move c4d4. Synthesis: no retrieved analog transferred to Qxf7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

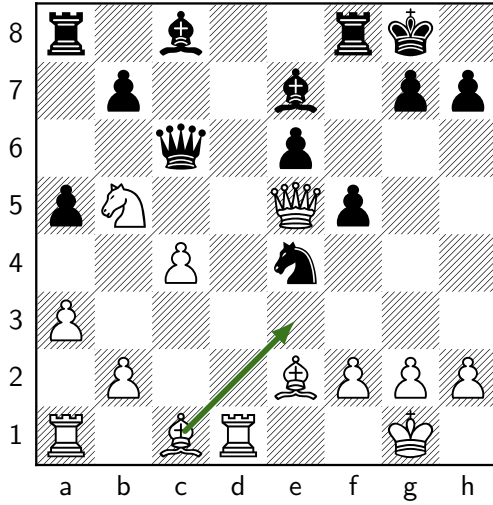




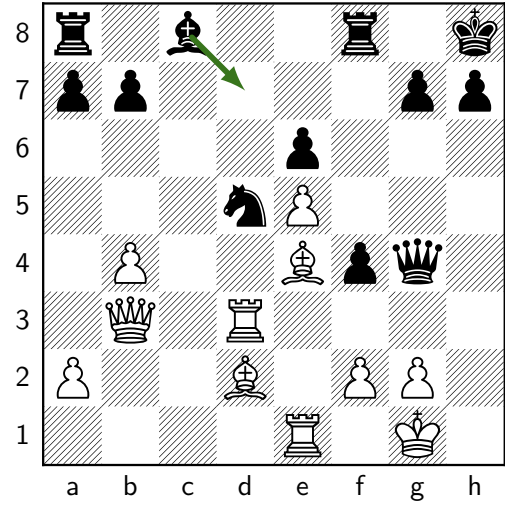
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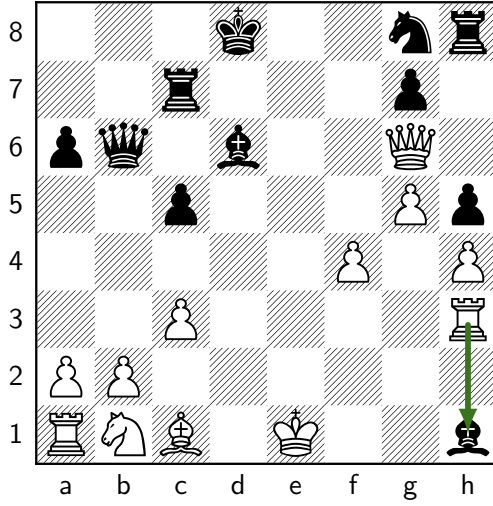
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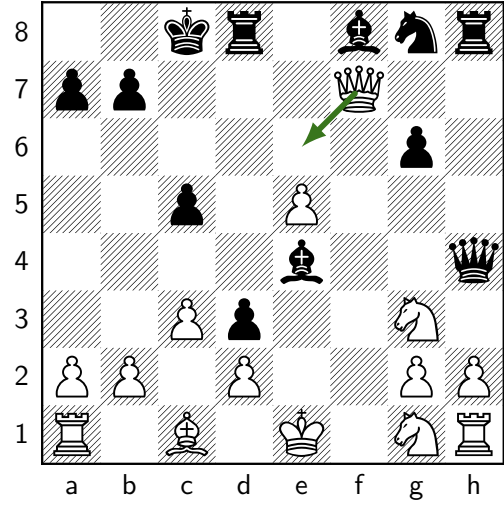
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Figure 15: **Move 21, White: Qg6.** (a) Our position; Melanie plays Qg6 (f7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3855` (ply 43) [structural distance  $d=119$ , QAP obj=76]: the master played Ne5; under  $\mu$  our Qf7 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Bh1 as their Pb7, our Qf7 as their Bd6; transferred move f7f6. (c) Retrieved from `ccr1_top_p0000#290` (ply 32) [structural distance  $d=108$ , QAP obj=48]: the master played Be3; under  $\mu$  our Pa2 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our ph5 as their Ra1, our qb6 as their Be2; transferred move a2a3. (d) Retrieved from `ccr1_top_p0002#2197` (ply 57) [structural distance  $d=105$ , QAP obj=45]: the master played Bd7; under  $\mu$  our Rh3 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our pc5 as their Ra8, our Ph4 as their Rf8; transferred move h3h2. Synthesis: no retrieved analog transferred to Qg6 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

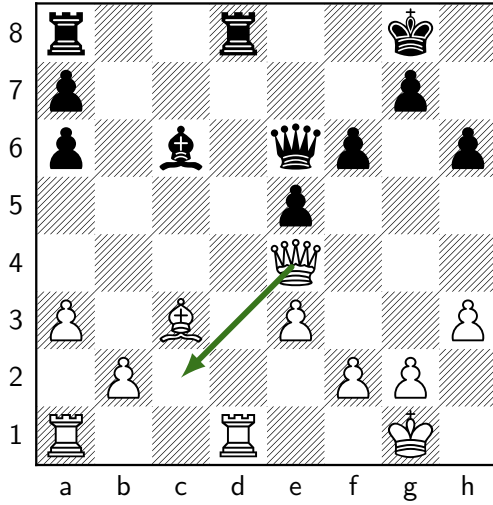




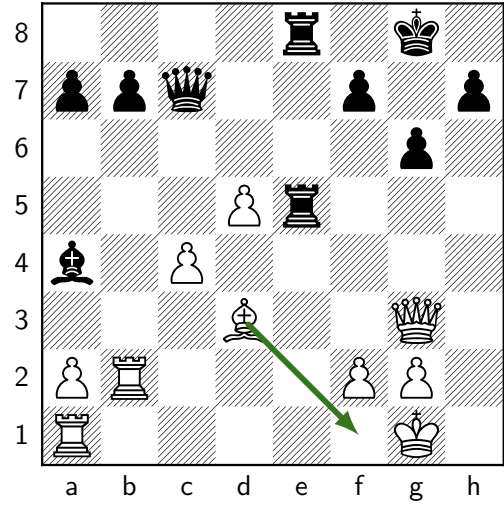
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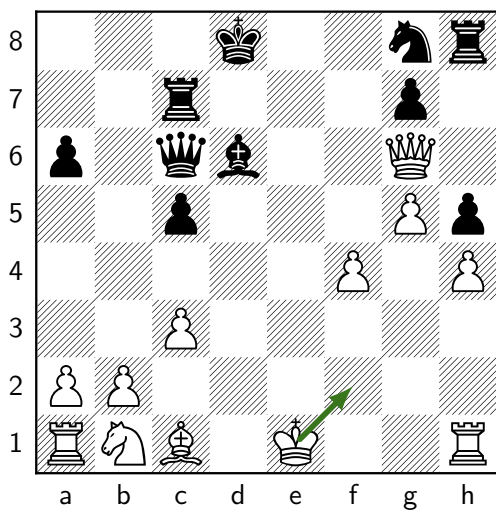


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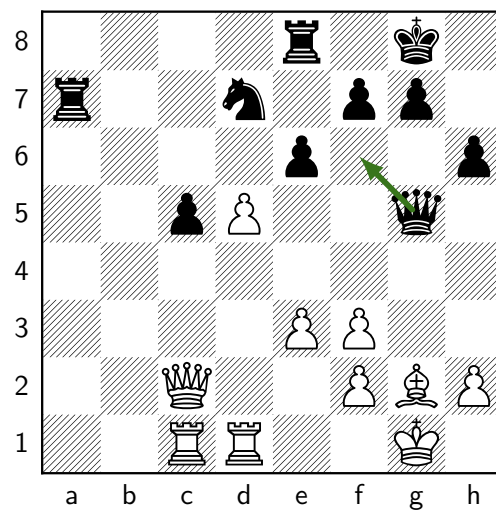


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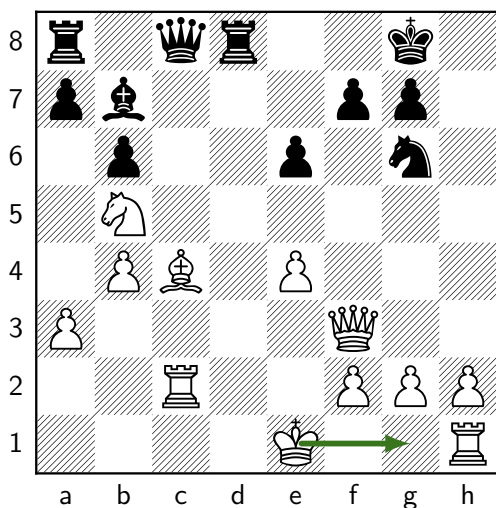
Figure 16: **Move 22, White: Rxh1.** (a) Our position; Melanie plays Rxh1 (h3h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#90` (ply 26) [structural distance  $d=91$ , QAP obj=67]: the master played Qe6+; under  $\mu$  our Rh3 plays the role of that game’s Qf7 (a **cross-type** correspondence); cross-type roles: our rh8 as their Bc1, our ph5 as their Ng1; transferred move h3e3. (c) Retrieved from `ccr1_top_p0000#3414` (ply 48) [structural distance  $d=100$ , QAP obj=57]: the master played Qc2; under  $\mu$  our Qg6 plays the role of that game’s Qe4; cross-type roles: our Bc1 as their Ra1, our pc5 as their Rd1; transferred move g6c2. (d) Retrieved from `ccr1_top_p0002#3519` (ply 44) [structural distance  $d=100$ , QAP obj=53]: the master played Bf1; under  $\mu$  our Bc1 plays the role of that game’s Bd3; cross-type roles: our Pc3 as their Rb2, our Nb1 as their Pf2; transferred move c1d2. Synthesis: no retrieved analog transferred to Rxh1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



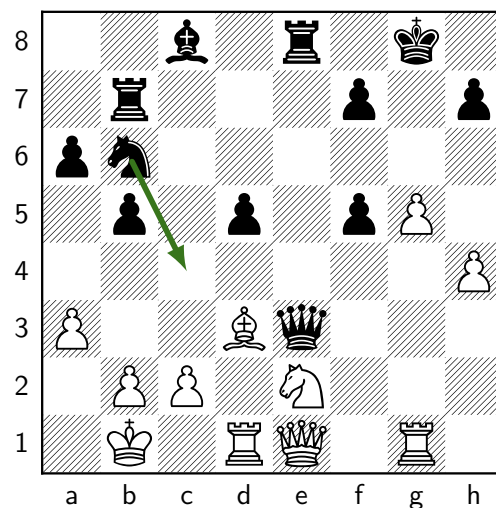
(a)



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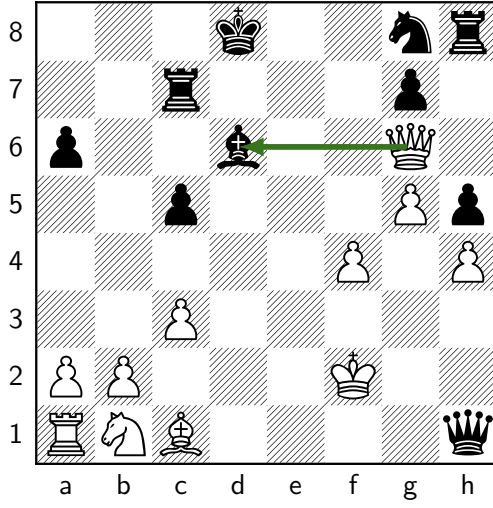


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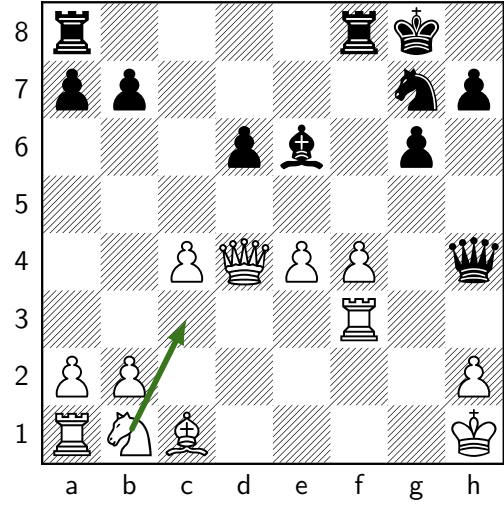


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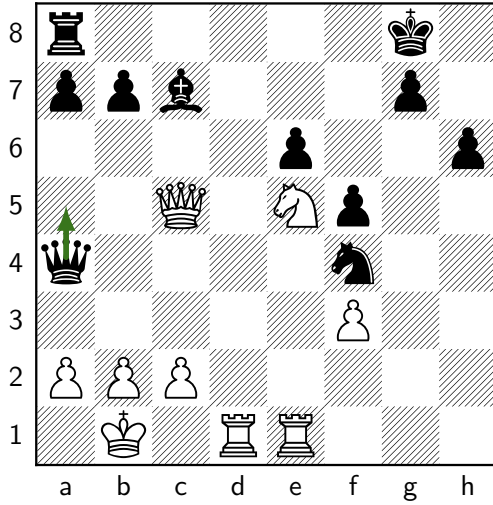
Figure 17: **Move 23, White: Kf2**. (a) Our position; Melanie plays Kf2 (e1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1022` (ply 47) [structural distance  $d=75$ , QAP obj=102]: the master played Qf6; under  $\mu$  our Qg6 plays the role of that game’s Qg4; cross-type roles: our Nb1 as their Re8, our Bc1 as their Nd7; transferred move g6f5. (c) Retrieved from `ccr1_top_p0000#2059` (ply 42) [structural distance  $d=84$ , QAP obj=93]: the master played 0-0; under  $\mu$  our Ke1 plays the role of that game’s Ke1; cross-type roles: our pa6 as their Rc2, our pc5 as their Qf3; transferred move e1d1. (d) Retrieved from `ccr1_top_p0002#2412` (ply 47) [structural distance  $d=94$ , QAP obj=63]: the master played Nc4; under  $\mu$  our Nb1 plays the role of that game’s Nb3; cross-type roles: our Ph4 as their Bc8, our Pg5 as their Re8; transferred move b1a3. Synthesis: no retrieved analog transferred to Kf2 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



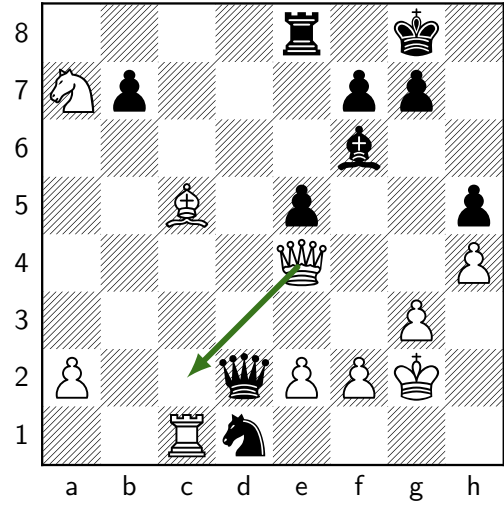
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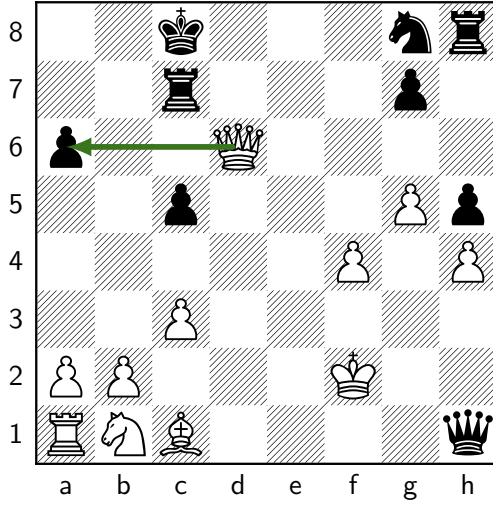


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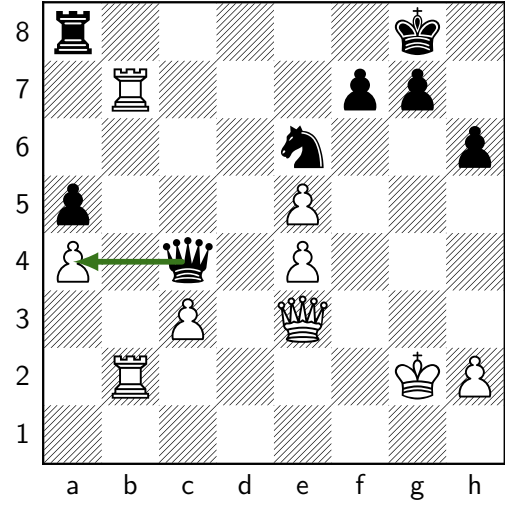


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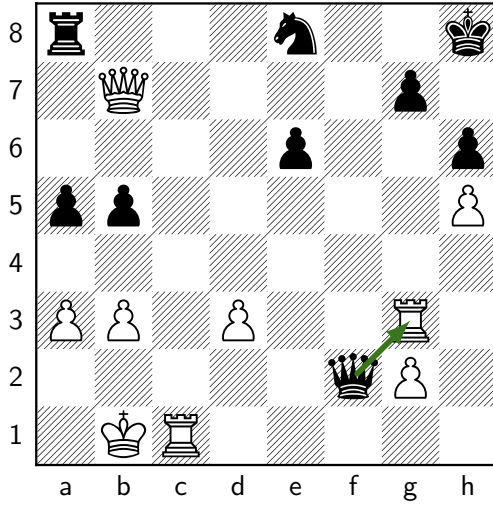
Figure 18: **Move 24, White: Qxd6+.** (a) Our position; Melanie plays Qxd6+ (g6d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#400` (ply 38) [structural distance  $d=87$ , QAP obj=73]: the master played Nc3; under  $\mu$  our Pc3 plays the role of that game's Nb1 (a **cross-type** correspondence); cross-type roles: our Pb2 as their Ra1, our Pf4 as their Bc1; transferred move c3c4. (c) Retrieved from `ccr1_top_p0001#1236` (ply 55) [structural distance  $d=78$ , QAP obj=58]: the master played Qa5; under  $\mu$  our Qg6 plays the role of that game's Qa5; cross-type roles: our ng8 as their Pb7, our Ra1 as their Pg7; transferred move g6h5. (d) Retrieved from `ccr1_top_p0002#1950` (ply 56) [structural distance  $d=92$ , QAP obj=31]: the master played Qc2; under  $\mu$  our Qg6 plays the role of that game's Qe4; cross-type roles: our Ph4 as their Rc1, our qh1 as their nd1; transferred move g6h5. Synthesis: no retrieved analog transferred to Qxd6+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



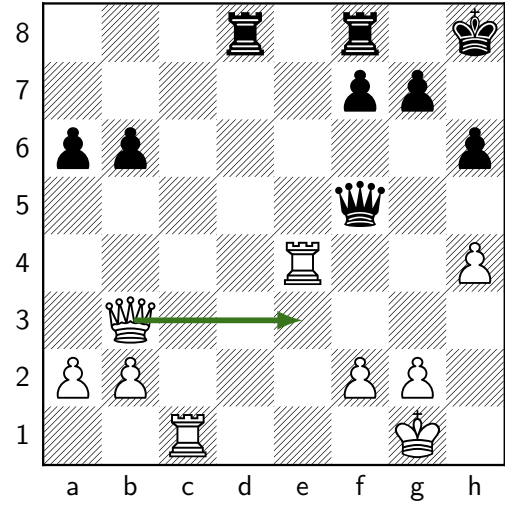
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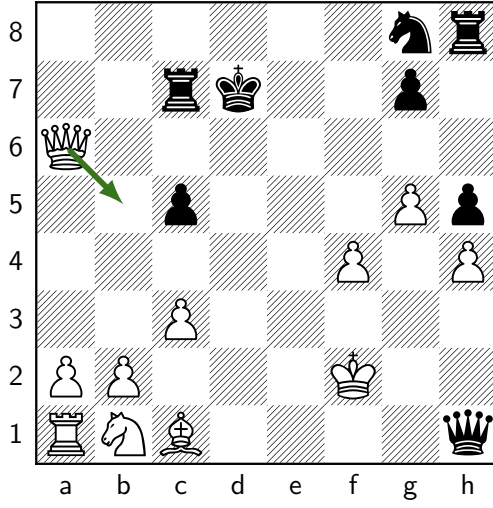


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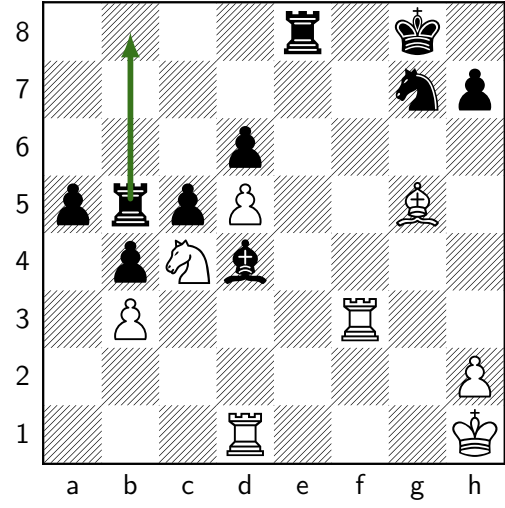


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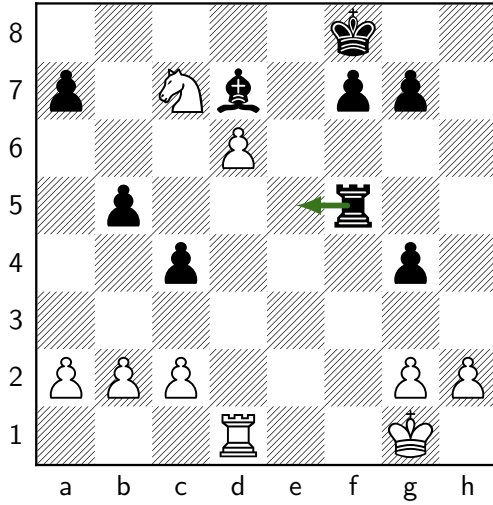
Figure 19: **Move 25, White: Qxa6+.** (a) Our position; Melanie plays Qxa6+ (d6a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1827` (ply 65) [structural distance  $d=94$ , QAP obj=3]: the master played Qxa4; under  $\mu$  our Qd6 plays the role of that game's Qc5; cross-type roles: our Pg5 as their Ra8, our qh1 as their rb7; transferred move d6a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#885` (ply 65) [structural distance  $d=90$ , QAP obj=35]: the master played Qxg3; under  $\mu$  our Qd6 plays the role of that game's Qf7; cross-type roles: our pg7 as their Ne8, our Ra1 as their Pa5; transferred move d6b6. (d) Retrieved from `ccr1_top_p0002#878` (ply 54) [structural distance  $d=96$ , QAP obj=27]: the master played Qe3; under  $\mu$  our Pc3 plays the role of that game's Qb3 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Rc1, our Ra1 as their Pa2; transferred move c3c4. Synthesis: 1 retrieved analog transferred to Qxa6+ (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



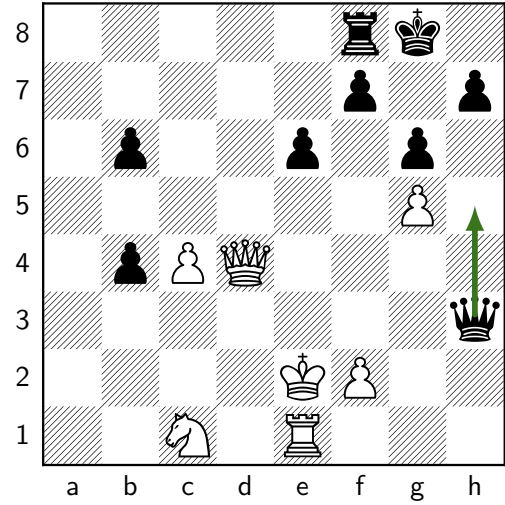
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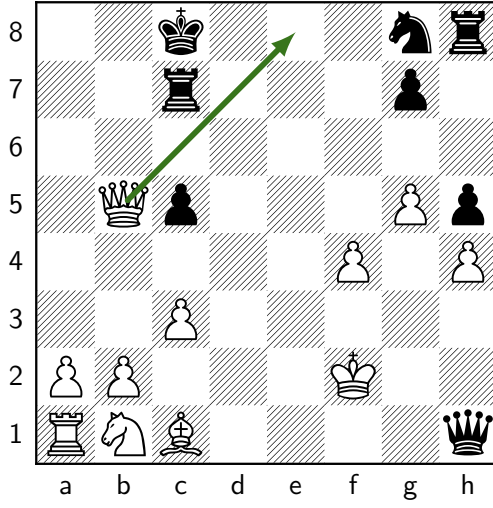


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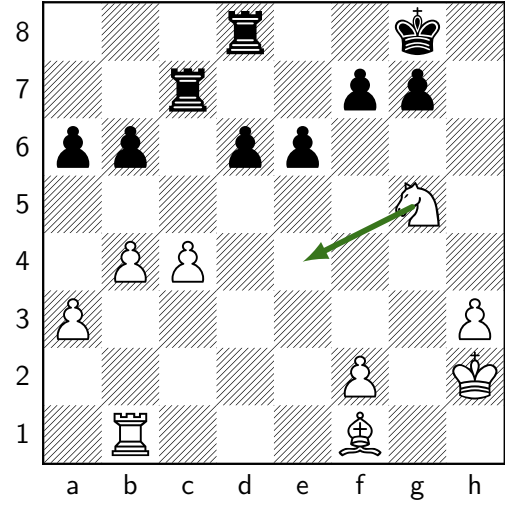


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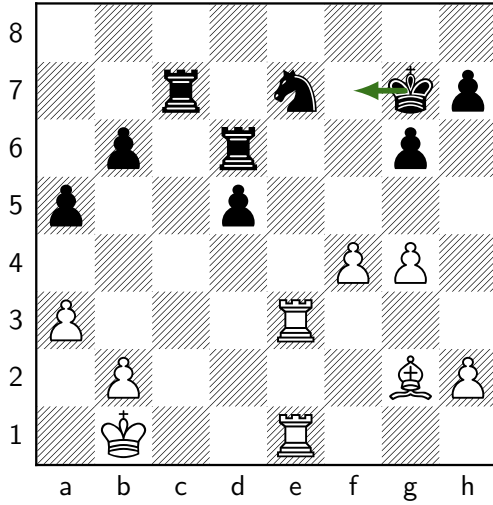
Figure 20: **Move 26, White: Qb5+.** (a) Our position; Melanie plays Qb5+ (a6b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3985` (ply 65) [structural distance  $d=83$ , QAP obj=19]: the master played Rbb8; under  $\mu$  our Qa6 plays the role of that game’s Rb4 (a **cross-type** correspondence); cross-type roles: our Pc3 as their Re8, our Pf4 as their Ng7; transferred move a6b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2570` (ply 59) [structural distance  $d=87$ , QAP obj=19]: the master played Re5; under  $\mu$  our Qa6 plays the role of that game’s Rf4 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Pa7, our qh1 as their nc7; transferred move a6c4. (d) Retrieved from `ccr1_top_p0003#345` (ply 63) [structural distance  $d=83$ , QAP obj=19]: the master played Qh5+; under  $\mu$  our Qa6 plays the role of that game’s Qh6; cross-type roles: our Nb1 as their Rf8, our Bc1 as their Pb6; transferred move a6a4. Synthesis: 1 retrieved analog transferred to Qb5+ (summed QAP objective 19, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



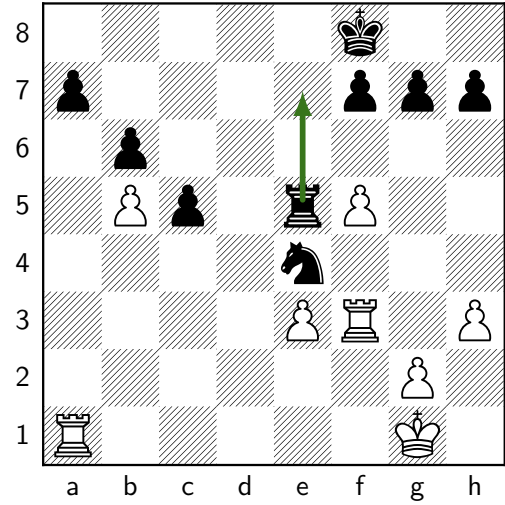
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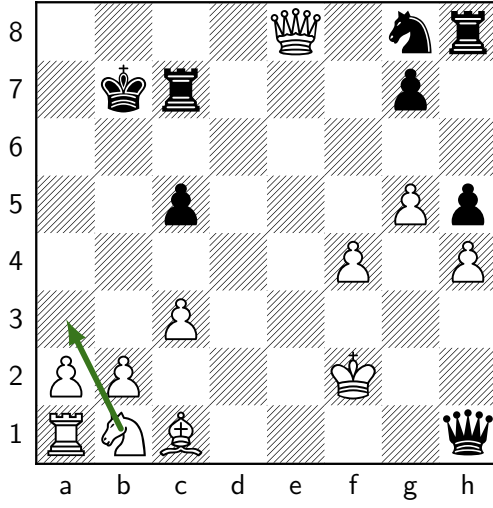


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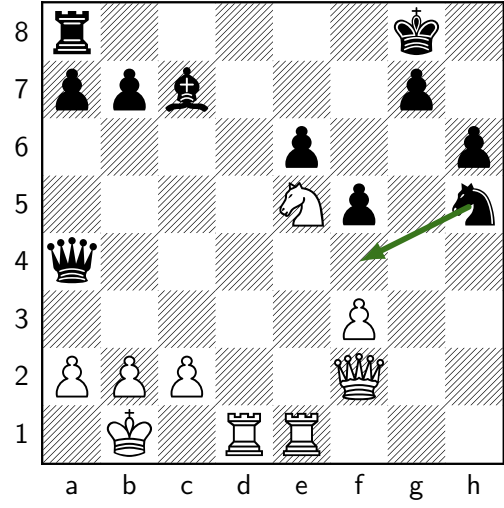


(d)

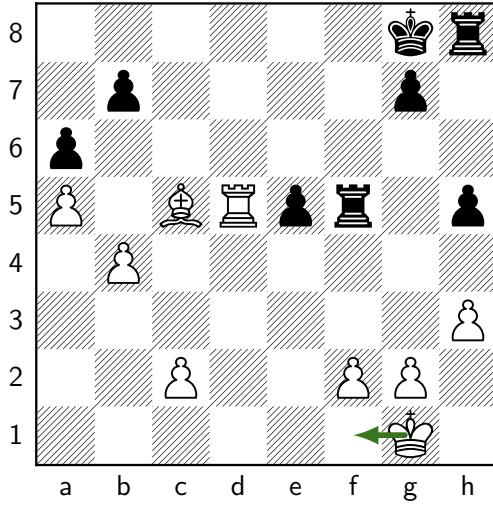
Figure 21: **Move 27, White: Qe8+.** (a) Our position; Melanie plays Qe8+ (b5e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#941` (ply 62) [structural distance  $d=77$ , QAP obj=26]: the master played Nc4; under  $\mu$  our Qb5 plays the role of that game's Ng5 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Rb1, our ng8 as their pb6; transferred move b5d3. (c) Retrieved from `ccr1_top_p0003#451` (ply 67) [structural distance  $d=83$ , QAP obj=24]: the master played Kf7; under  $\mu$  our Kf2 plays the role of that game's Kg2; cross-type roles: our pc5 as their Ne7, our Nb1 as their Rd6; transferred move f2g3. (d) Retrieved from `ccr1_top_p0000#1262` (ply 67) [structural distance  $d=78$ , QAP obj=23]: the master played Re7; under  $\mu$  our Qb5 plays the role of that game's Re4 (a **cross-type** correspondence); cross-type roles: our qh1 as their Pg7, our Ra1 as their Ph7; transferred move b5f1. Synthesis: no retrieved analog transferred to Qe8+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



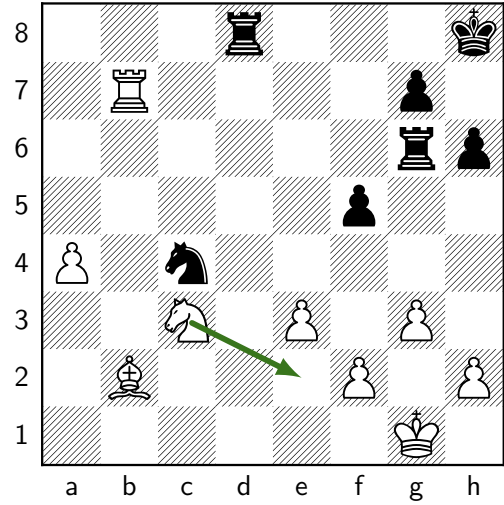
(a)



(b)

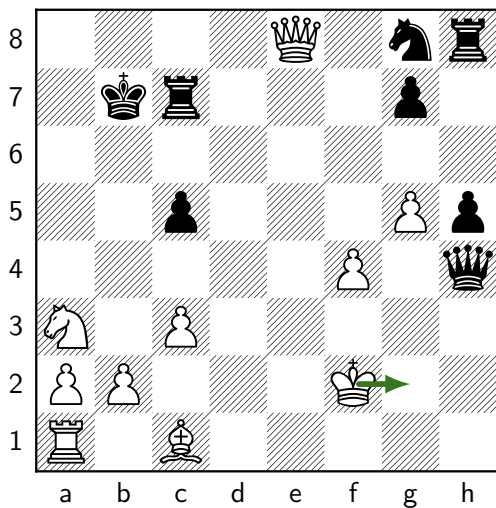


(c)

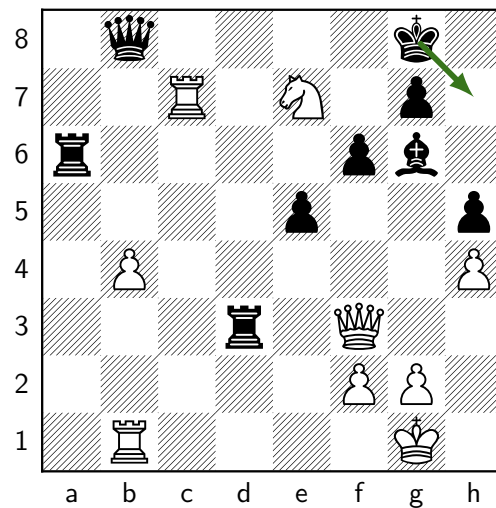


(d)

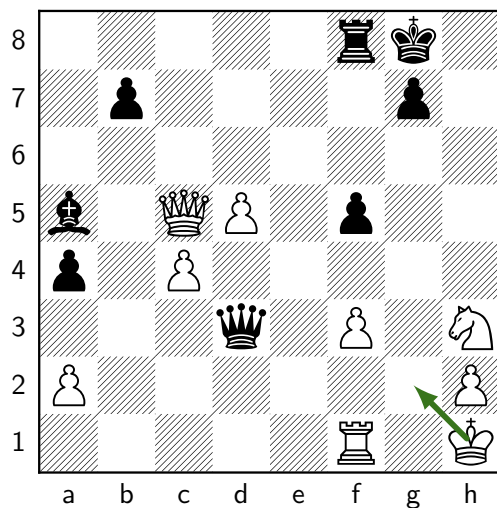
Figure 22: **Move 28, White: Na3.** (a) Our position; Melanie plays Na3 (b1a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1236` (ply 53) [structural distance  $d=78$ , QAP obj=31]: the master played Nf4; under  $\mu$  our Nb1 plays the role of that game’s Nh4; cross-type roles: our Bc1 as their Ra8, our Qe8 as their Bc7; transferred move b1a3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#74` (ply 66) [structural distance  $d=83$ , QAP obj=34]: the master played Kf1; under  $\mu$  our Kf2 plays the role of that game’s Kg1; cross-type roles: our Nb1 as their Pc2, our Bc1 as their Pf2; transferred move f2g3. (d) Retrieved from `ccrl_top_p0000#575` (ply 64) [structural distance  $d=84$ , QAP obj=32]: the master played Ne2; under  $\mu$  our Pc3 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our ng8 as their Bb2, our Bc1 as their Pg3; transferred move c3c4. Synthesis: 1 retrieved analog transferred to Na3 (summed QAP objective 31, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)



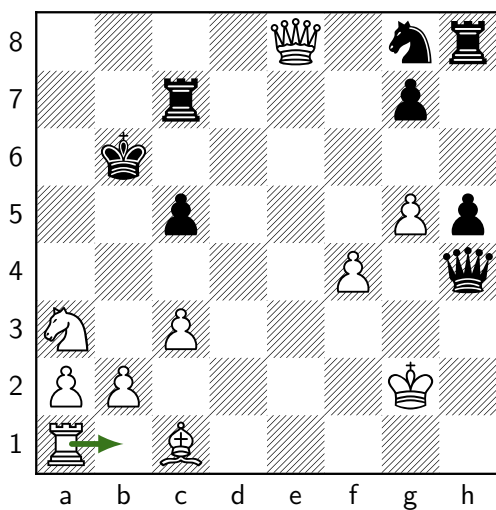
(b)



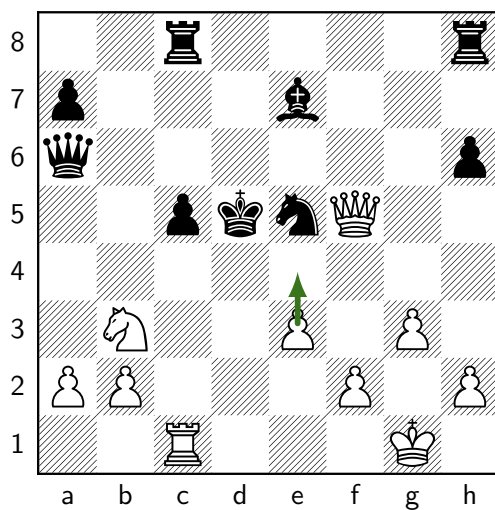
(c)

Figure 23: **Move 29, White: Kg2.** (a) Our position; Melanie plays Kg2 (f2g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0000#206` (ply 73) [structural distance  $d=81$ , QAP obj=69]: the master played Kh7; under  $\mu$  our Kf2 plays the role of that game’s Kg1; cross-type roles: our qh4 as their ne7, our Bc1 as their Pg7; transferred move f2g2 (**endorses this move**). (c) Retrieved from `ccr1_top-p0002#303` (ply 64) [structural distance  $d=84$ , QAP obj=11]: the master played Kg2; under  $\mu$  our Kf2 plays the role of that game’s Kh1; cross-type roles: our pg7 as their Rf1, our Na3 as their Pa2; transferred move f2g2 (**endorses this move**). Synthesis: 2 retrieved analogs transferred to Kg2 (summed QAP objective 81, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

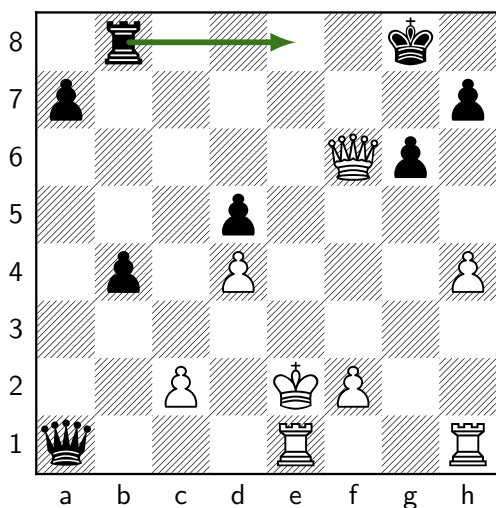




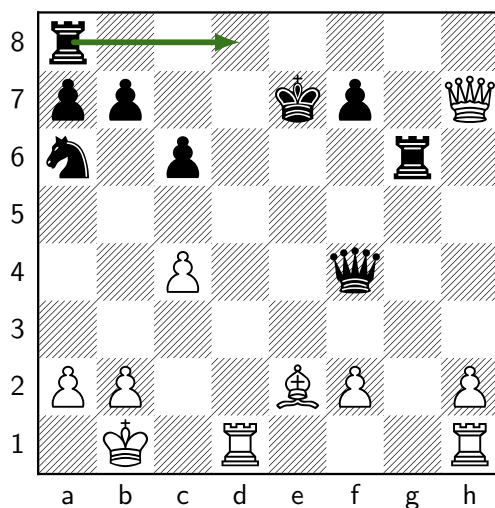
(a)



(b)

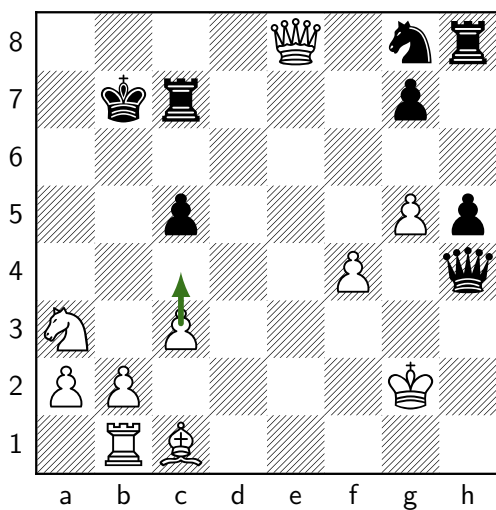


(c)

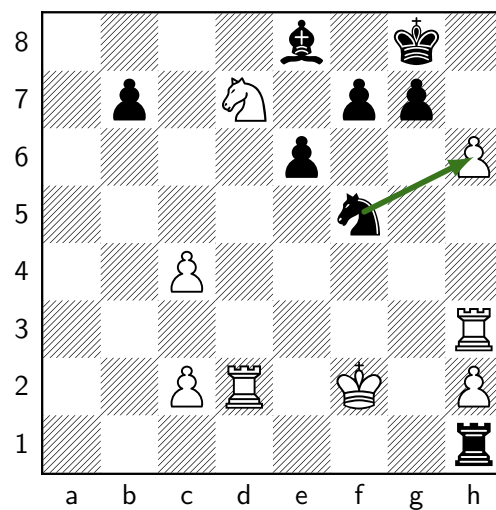


(d)

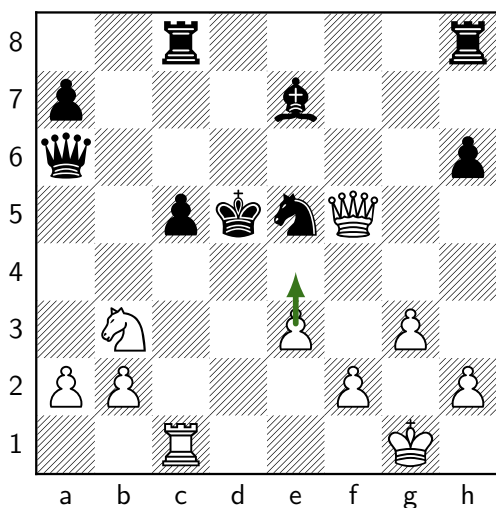
Figure 24: **Move 30, White: Rb1**. (a) Our position; Melanie plays Rb1 (a1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3789` (ply 46) [structural distance  $d=79$ , QAP obj=26]: the master played e4+; under  $\mu$  our *Ra1* plays the role of that game’s *Pe3* (a **cross-type** correspondence); cross-type roles: our *Pg5* as their *Rc1*, our *rc7* as their *Pa2*; transferred move a1b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#80` (ply 73) [structural distance  $d=74$ , QAP obj=22]: the master played Re8+; under  $\mu$  our *Ra1* plays the role of that game’s *Rb1*; cross-type roles: our *Bc1* as their *Pa7*, our *Na3* as their *pd4*; transferred move a1b1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1946` (ply 53) [structural distance  $d=77$ , QAP obj=12]: the master played Rd8; under  $\mu$  our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *ng8* as their *Pf7*, our *ph5* as their *Rg6*; transferred move a1b1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rb1 (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



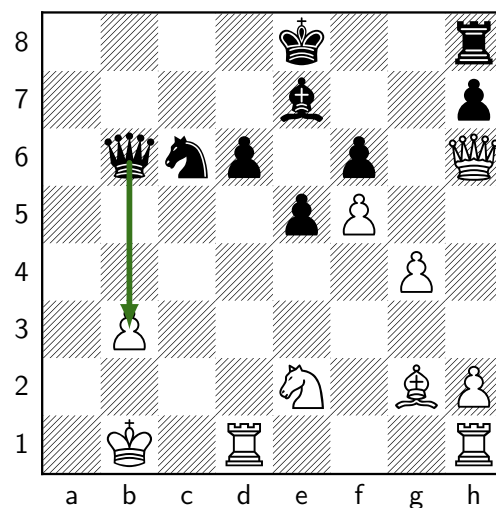
(a)



(b)

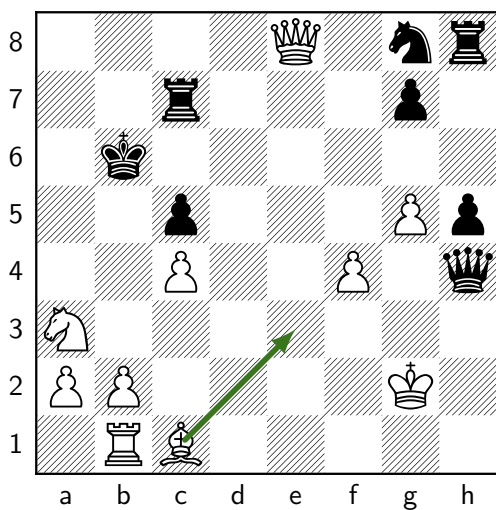


(c)

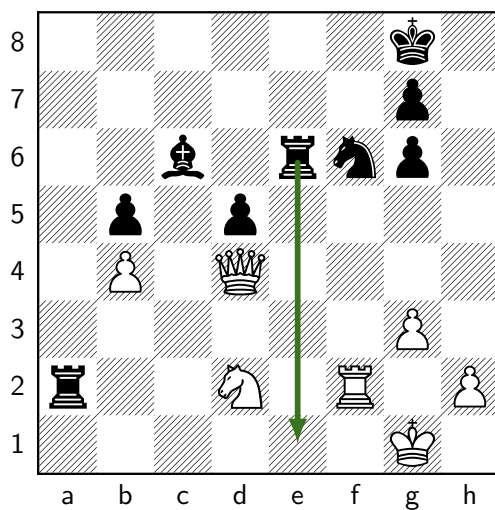


(d)

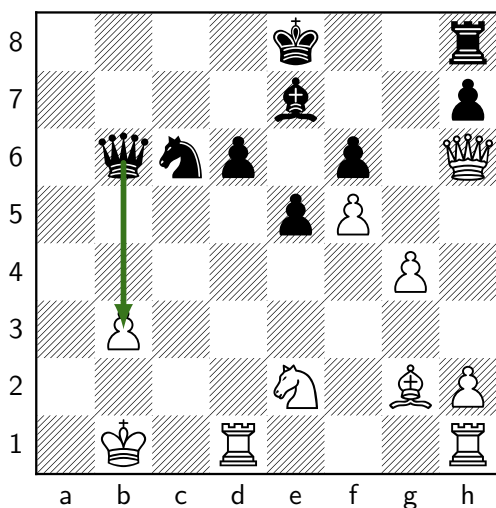
Figure 25: **Move 31, White: c4**. (a) Our position; Melanie plays c4 (c3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0000#2614` (ply 77) [structural distance  $d=81$ , QAP obj=31]: the master played  $Nxh6$ ; under  $\mu$  our  $Pc3$  plays the role of that game’s  $Nf4$  (a **cross-type** correspondence); cross-type roles: our  $qh4$  as their  $Be8$ , our  $Pg5$  as their  $nd7$ ; transferred move  $c3c4$  (**endorses this move**). (c) Retrieved from `ccr1.top_p0000#3789` (ply 46) [structural distance  $d=80$ , QAP obj=27]: the master played  $e4+$ ; under  $\mu$  our  $Pc3$  plays the role of that game’s  $Pe3$ ; cross-type roles: our  $Pg5$  as their  $Rc1$ , our  $rc7$  as their  $Pa2$ ; transferred move  $c3c4$  (**endorses this move**). (d) Retrieved from `ccr1.top_p0001#1119` (ply 47) [structural distance  $d=63$ , QAP obj=43]: the master played  $Qxb3+$ ; under  $\mu$  our  $Qe8$  plays the role of that game’s  $Qb3$ ; cross-type roles: our  $pg7$  as their  $Be7$ , our  $ng8$  as their  $Ph7$ ; transferred move  $e8g6$ . Synthesis: 2 retrieved analogs transferred to c4 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



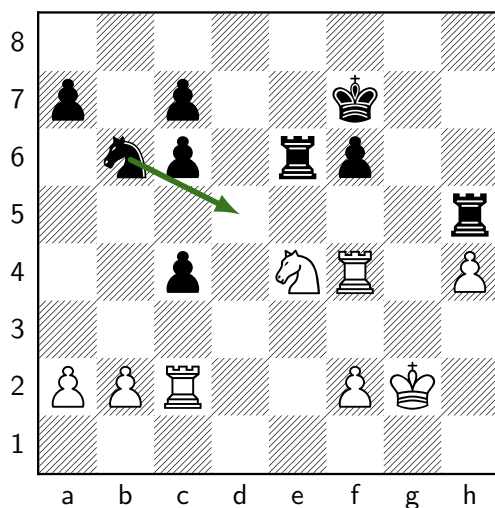
(a)



(b)



(c)



(d)

Figure 26: **Move 32, White: Be3.** (a) Our position; Melanie plays Be3 (c1e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2025` (ply 61) [structural distance  $d=74$ , QAP obj=55]: the master played Re1+; under  $\mu$  our Bc1 plays the role of that game’s Re3 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Bc6, our Pg5 as their Nf6; transferred move c1e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1119` (ply 47) [structural distance  $d=72$ , QAP obj=42]: the master played Qxb3+; under  $\mu$  our Qe8 plays the role of that game’s Qb3; cross-type roles: our pg7 as their Be7, our ng8 as their Ph7; transferred move e8g6. (d) Retrieved from `ccr1_top_p0002#1688` (ply 69) [structural distance  $d=80$ , QAP obj=36]: the master played Nd5; under  $\mu$  our Na3 plays the role of that game’s Nb3; cross-type roles: our Bc1 as their Pc6, our Qe8 as their Re6; transferred move a3c2. Synthesis: 1 retrieved analog transferred to Be3 (summed QAP objective 55, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

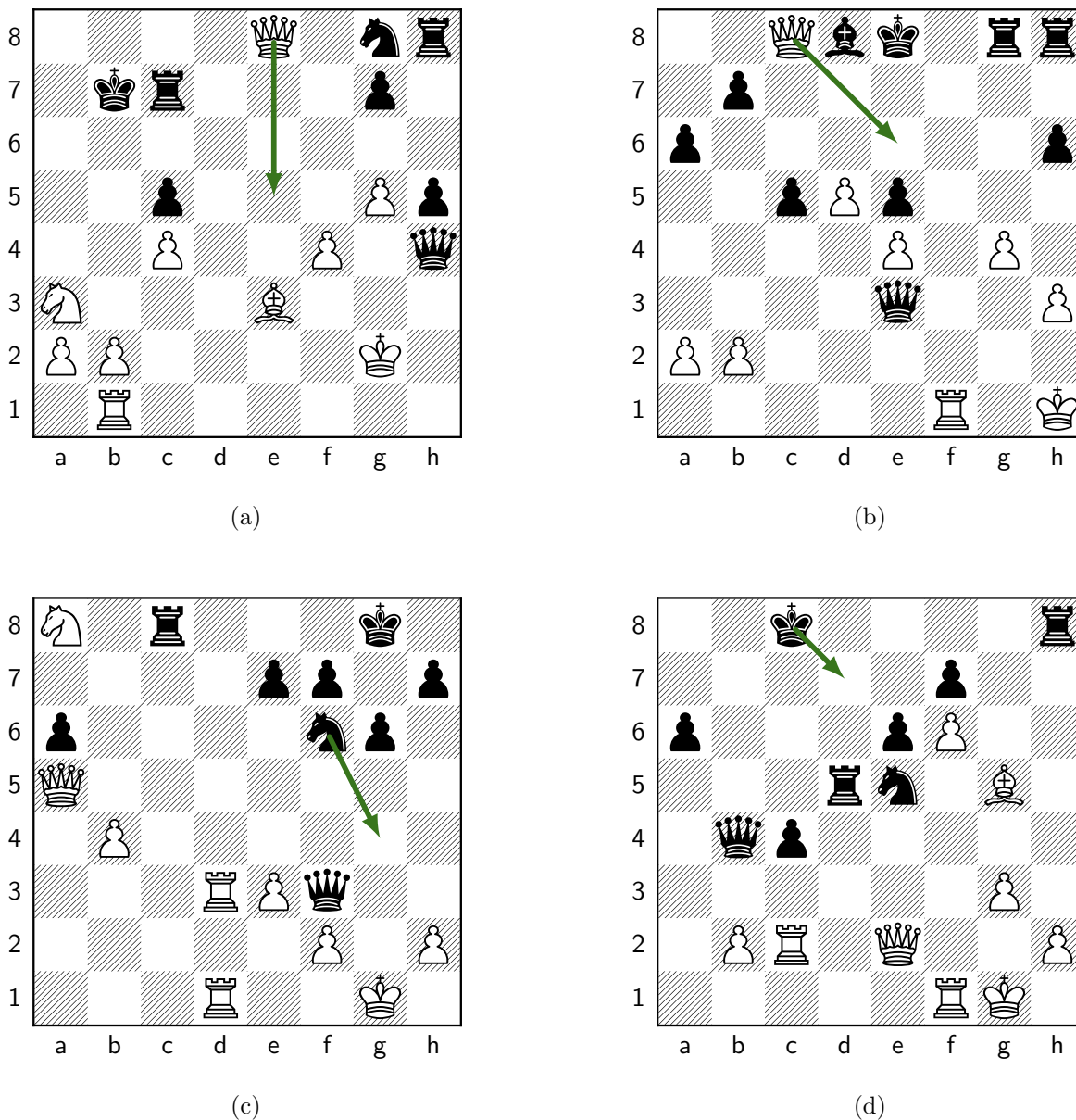
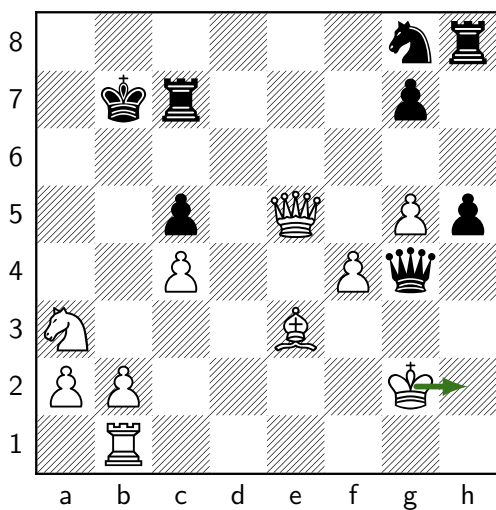
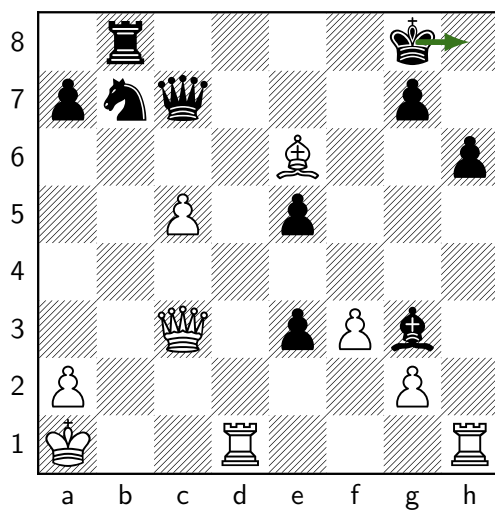


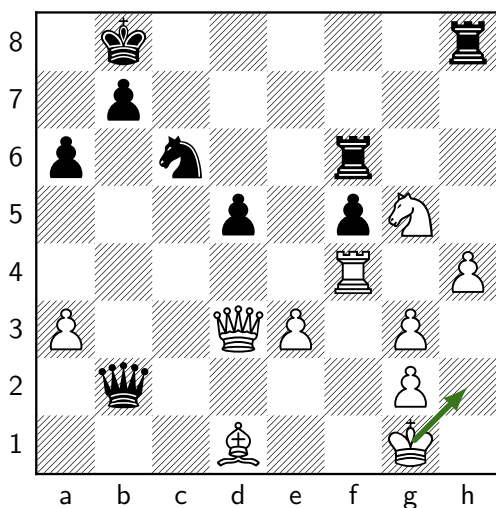
Figure 27: **Move 33, White: Qe5.** (a) Our position; Melanie plays Qe5 (e8e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3996` (ply 66) [structural distance  $d=72$ , QAP obj=38]: the master played Qe6+; under  $\mu$  our Qe8 plays the role of that game’s Qc8; cross-type roles: our Pa2 as their Rf1, our Na3 as their Pb2; transferred move e8e5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3819` (ply 61) [structural distance  $d=70$ , QAP obj=50]: the master played Ng4; under  $\mu$  our Be3 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Qe8 as their Rc8, our Na3 as their Pg6; transferred move e3d4. (d) Retrieved from `ccrl_top_p0002#1426` (ply 51) [structural distance  $d=72$ , QAP obj=35]: the master played Kd7; under  $\mu$  our Kg2 plays the role of that game’s Kc1; cross-type roles: our pc5 as their Rh8, our Rb1 as their Pa6; transferred move g2g1. Synthesis: 2 retrieved analogs transferred to Qe5 (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



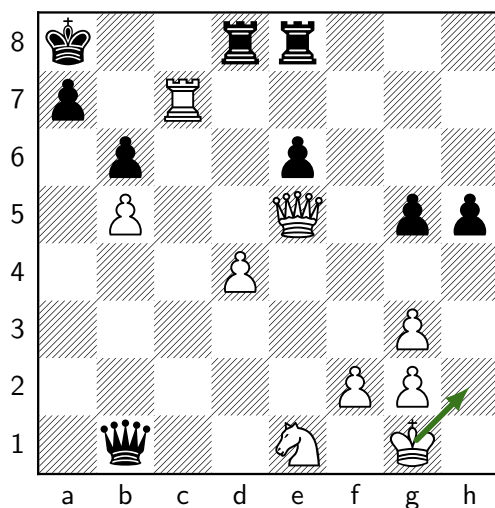
(a)



(b)

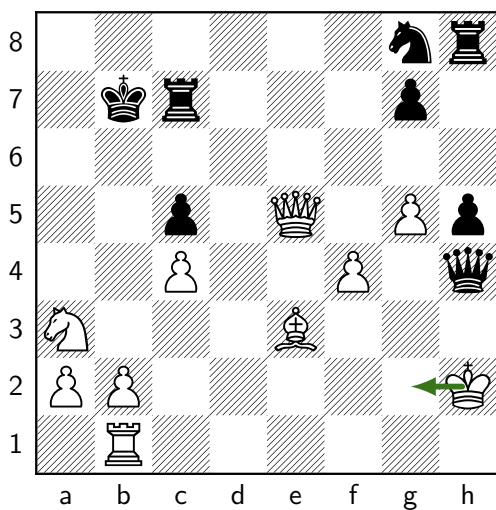


(c)

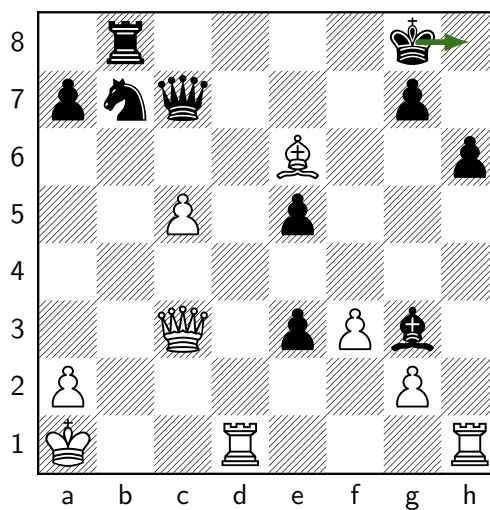


(d)

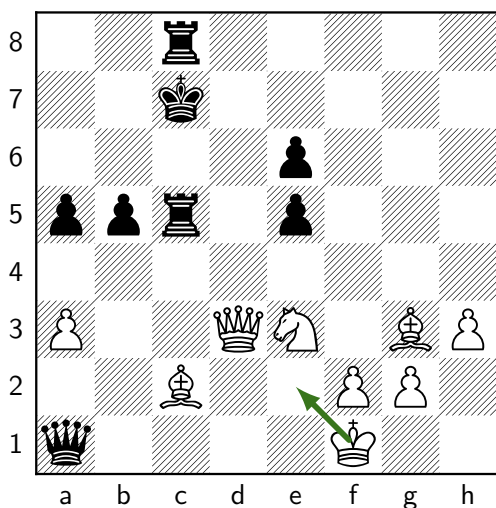
Figure 28: **Move 34, White: Kh2.** (a) Our position; Melanie plays Kh2 (g2h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3745` (ply 55) [structural distance  $d=85$ , QAP obj=96]: the master played Kh8; under  $\mu$  our Kg2 plays the role of that game’s Kg1; cross-type roles: our Be3 as their Nb7, our qg4 as their be6; transferred move g2h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1913` (ply 62) [structural distance  $d=88$ , QAP obj=18]: the master played Kh2; under  $\mu$  our Kg2 plays the role of that game’s Kg1; cross-type roles: our Pg5 as their Bd1, our Be3 as their Qd3; transferred move g2h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1414` (ply 70) [structural distance  $d=82$ , QAP obj=15]: the master played Kh2; under  $\mu$  our Kg2 plays the role of that game’s Kg1; cross-type roles: our pg7 as their Ne1, our Rb1 as their Pg2; transferred move g2h2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kh2 (summed QAP objective 128, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



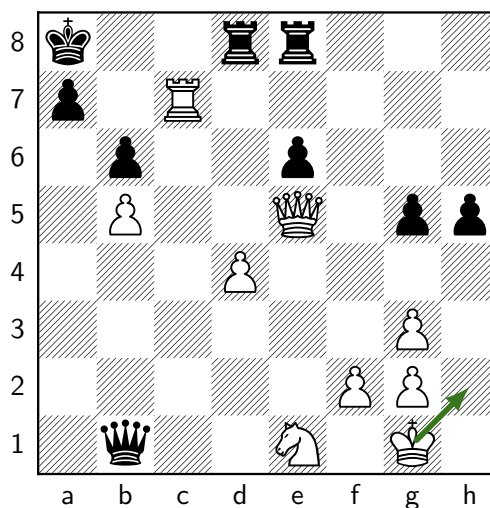
(a)



(b)

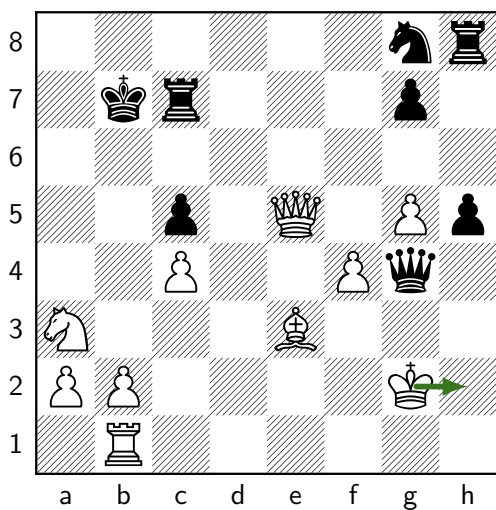


(c)

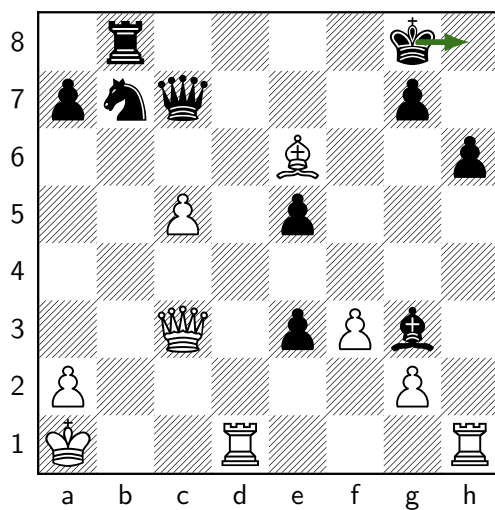


(d)

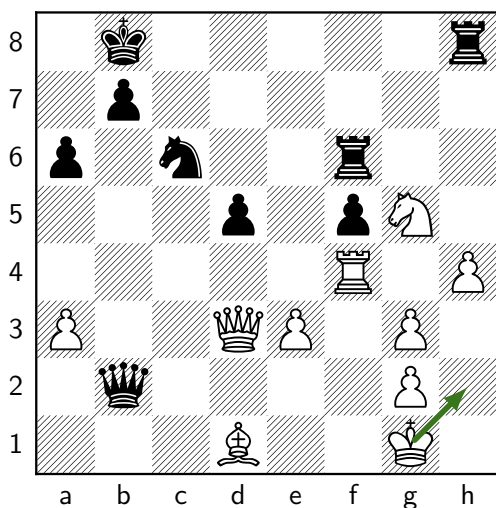
Figure 29: **Move 35, White: Kg2.** (a) Our position; Melanie plays Kg2 (h2g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3745` (ply 55) [structural distance  $d=87$ , QAP obj=96]: the master played Kh8; under  $\mu$  our Kh2 plays the role of that game’s Kg1; cross-type roles: our Be3 as their Nb7, our qh4 as their be6; transferred move h2g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1172` (ply 80) [structural distance  $d=90$ , QAP obj=53]: the master played Ke2; under  $\mu$  our Kh2 plays the role of that game’s Kf1; cross-type roles: our ph5 as their Qd3, our Pg5 as their Ne3; transferred move h2g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1414` (ply 70) [structural distance  $d=83$ , QAP obj=15]: the master played Kh2; under  $\mu$  our Kh2 plays the role of that game’s Kg1; cross-type roles: our pg7 as their Ne1, our Rb1 as their Pg2; transferred move h2g2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg2 (summed QAP objective 164, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



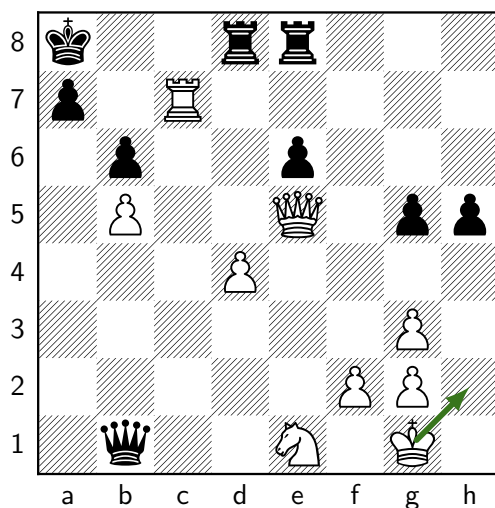
(a)



(b)



(c)



(d)

Figure 30: **Move 36, White: Kh2.** (a) Our position; Melanie plays Kh2 (g2h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3745` (ply 55) [structural distance  $d=85$ , QAP obj=96]: the master played Kh8; under  $\mu$  our Kg2 plays the role of that game’s Kg1; cross-type roles: our Be3 as their Nb7, our qg4 as their be6; transferred move g2h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1913` (ply 62) [structural distance  $d=88$ , QAP obj=18]: the master played Kh2; under  $\mu$  our Kg2 plays the role of that game’s Kg1; cross-type roles: our Pg5 as their Bd1, our Be3 as their Qd3; transferred move g2h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1414` (ply 70) [structural distance  $d=82$ , QAP obj=15]: the master played Kh2; under  $\mu$  our Kg2 plays the role of that game’s Kg1; cross-type roles: our pg7 as their Ne1, our Rb1 as their Pg2; transferred move g2h2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kh2 (summed QAP objective 128, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.